Student Enrollment & Course Management System (SECMS)

Introduction

1.1 Project Overview

The Student Enrollment & Course Management System (SECMS) is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. It enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting — all within a unified Salesforce environment.

This project centralizes all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. Enrollment Officers can manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization ensures that all enrollment operations — from admission to course completion — are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

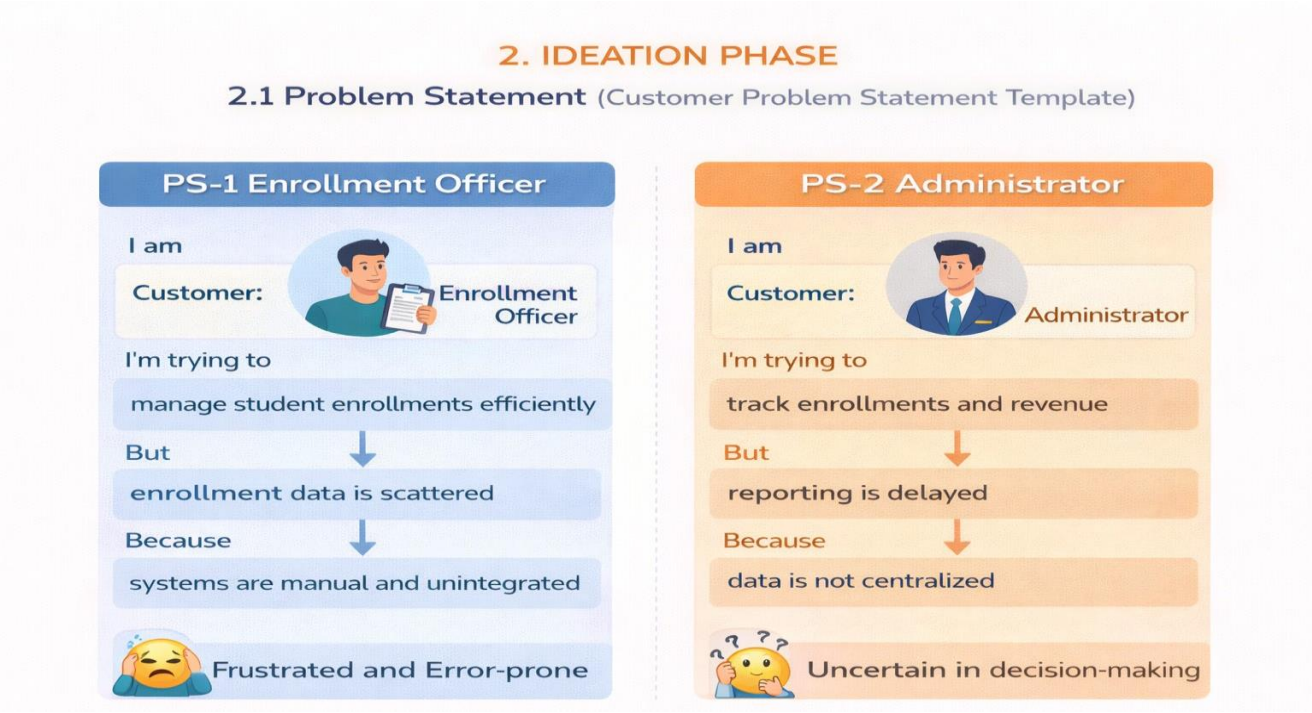
- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring.

2. Ideation Phase

2.1 Problem Statement

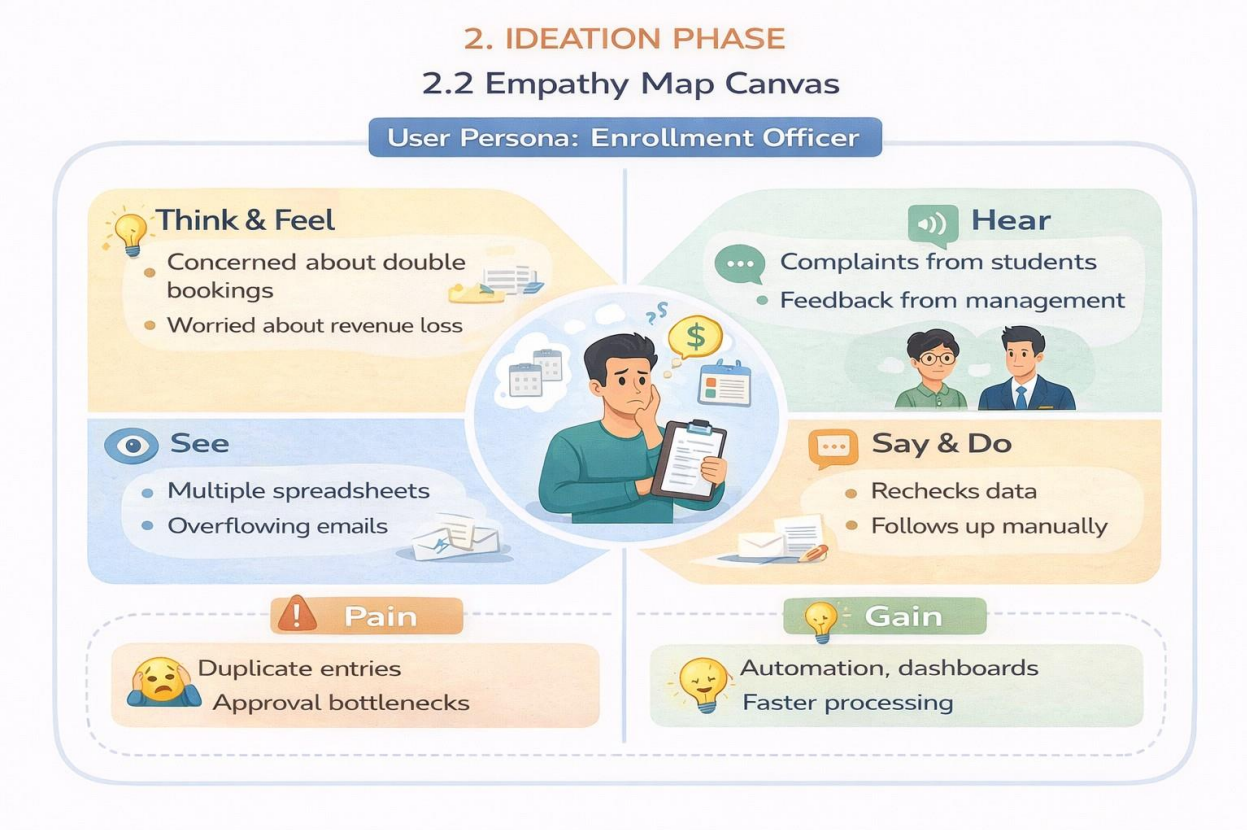
PS-1	Enrollment Officer	manage student enrollments efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone

PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralized	uncertain in decision-making



2.2 Empathy Map Canvas — User Persona: Enrollment Officer

Think & Feel	Concerned about double bookings and revenue loss
Hear	Complaints from students; feedback from management
See	Multiple spreadsheets and overflowing emails
Say & Do	Rechecks data; follows up manually
Pain	Duplicate entries, approval bottlenecks
Gain	Automation, dashboards, faster processing



2.3 Brainstorming & Idea Prioritization

Step 1 — Brainstormed Ideas

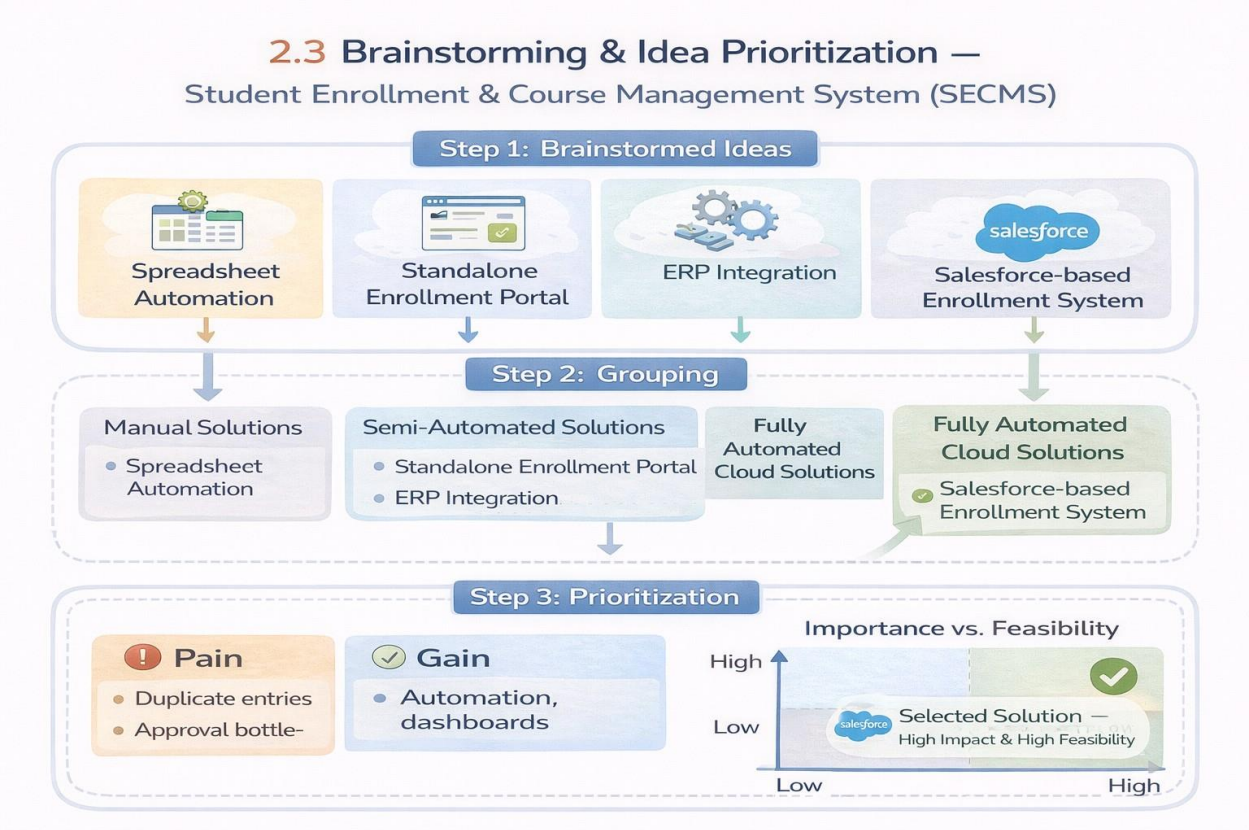
- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 — Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 — Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.

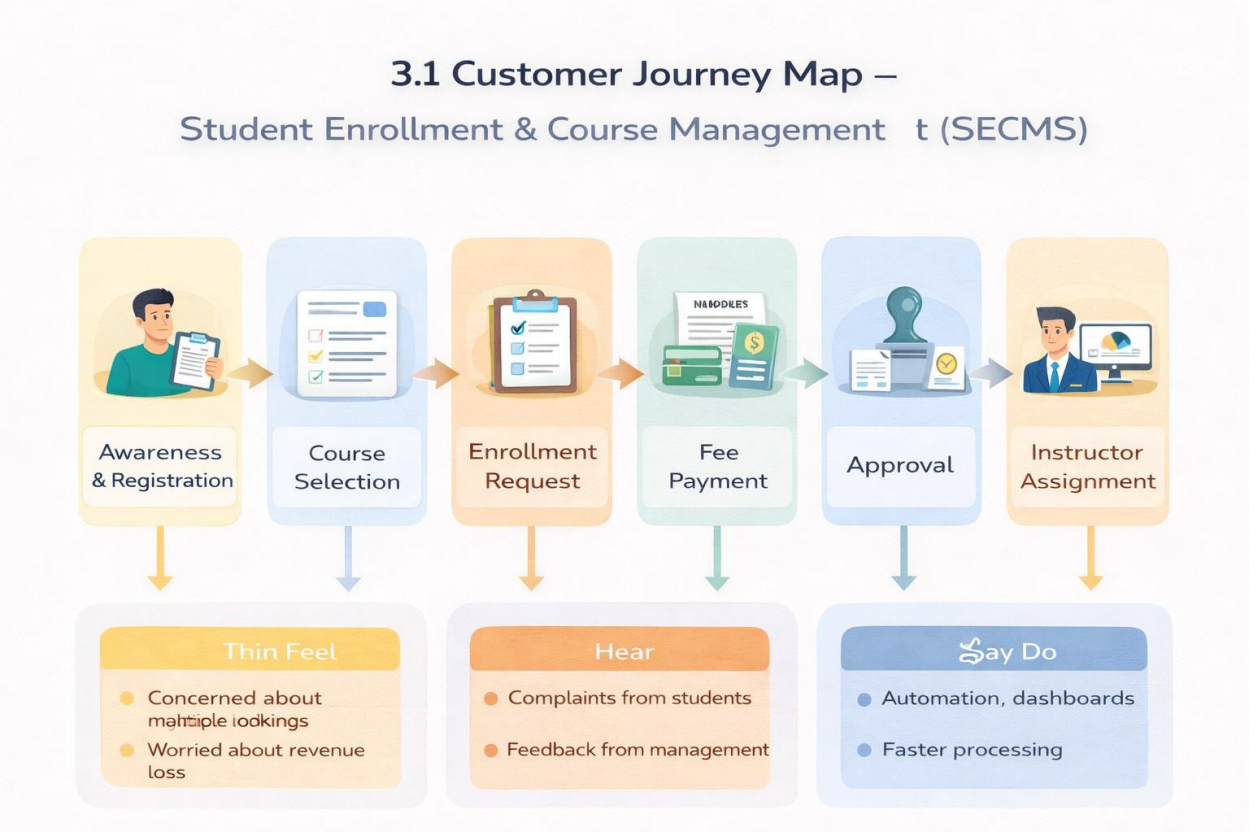


3. Requirement Analysis Phase

3.1 Customer Journey Map

Stages covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion



3.2 Solution Requirements

Functional Requirements

FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information shall only be accessible to authorized users.

- Only admins can modify course details
- Instructors view only assigned students

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

- No duplicate course registration
- Accurate status updates after approvals
- Consistent report generation

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality.

- Add new departments without redesign
- Handle increased admissions annually
- Extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime, supporting high availability for academic operations.

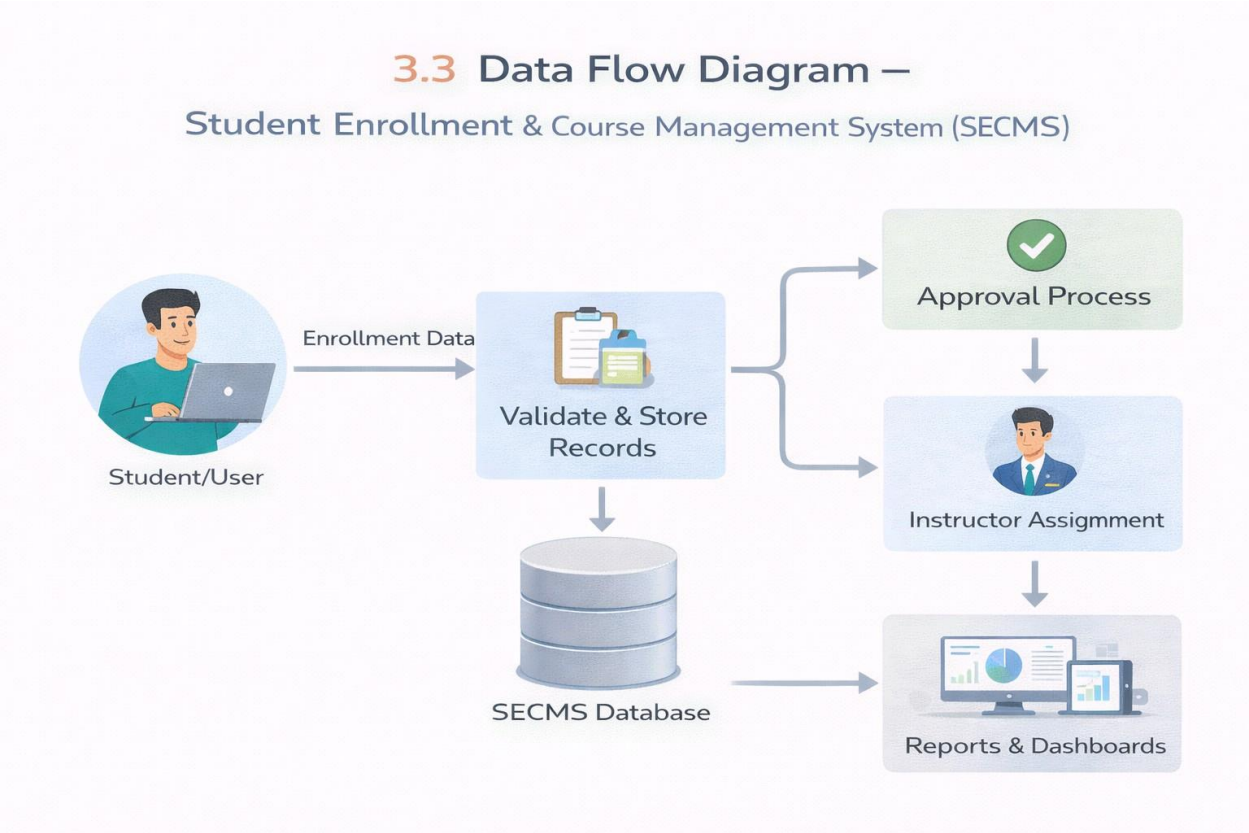
- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

Flow: Student/User → Enrollment Data → Validate & Store Records → SECMS Database; Validate & Store Records → Approval Process → Instructor Assignment → Reports & Dashboards.



3.4 Technology Stack

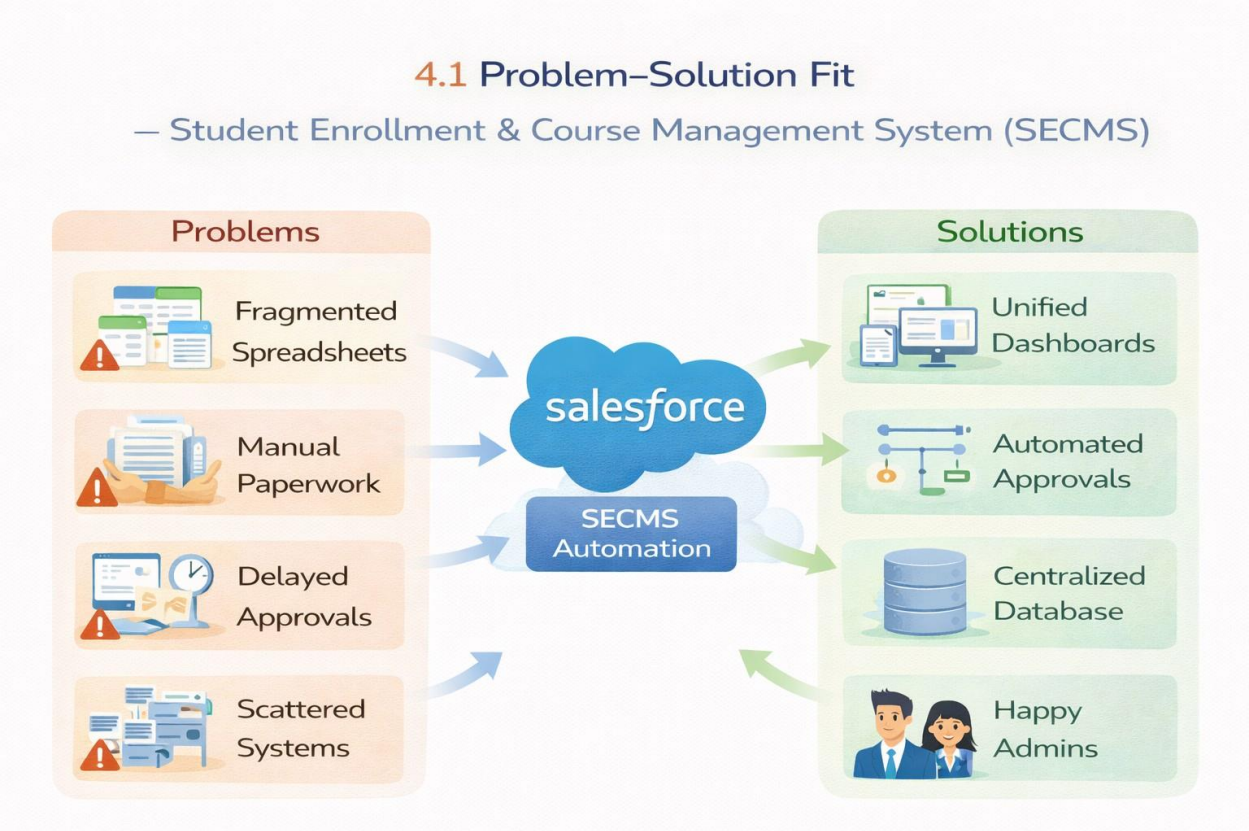
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

Fragmented spreadsheets	Unified dashboards
Manual paperwork	Automated approvals
Delayed approvals	Centralized database
Scattered systems	Happy admins



4.2 Proposed Solutions

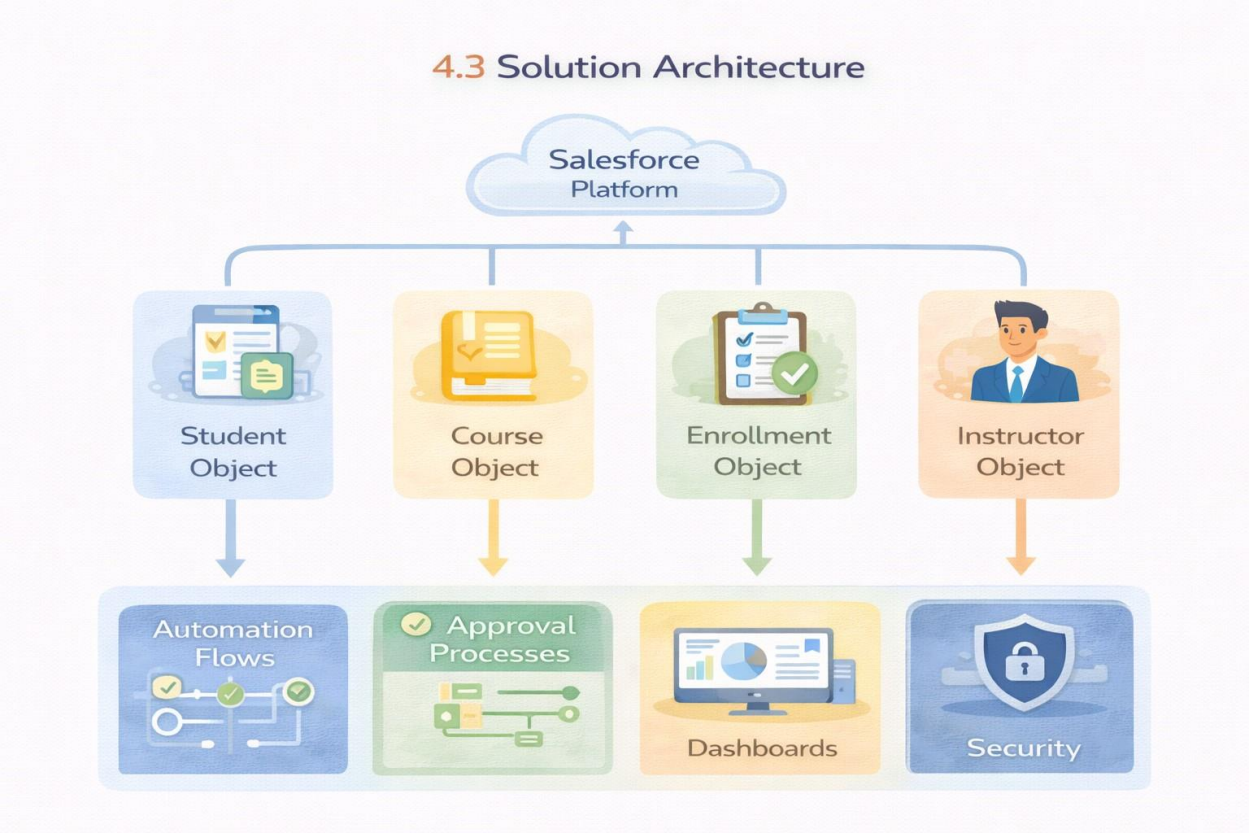
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience

Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.

Objects	Student Object, Course Object, Enrollment Object, Instructor Object
Processing	Automation Flows, Approval Processes
Insights & Access	Dashboards, Security



5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog & Sprint Schedule

Sprint-1	Developer Setup	USN-1	As a system admin, I want to create a Salesforce developer account so I can configure and deploy the Student Enrollment system.	1	High	Member 1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so academic data can be stored and managed efficiently.	3	High	Member 2
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member 3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so complete records can be captured accurately.	5	High	Member 4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so I can access all academic operations from one interface.	3	High	Member 4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so the system can logically link and track academic activities.	5	Medium	Member 1
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry.	5	High	Member 2
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so incorrect or incomplete records are prevented.	3	High	Member 4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so manual effort is reduced and processes become faster.	3	High	Member 4

Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so registrations are verified before confirmation.	5	High	Member 1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so users can only access data relevant to their responsibilities.	5	High	Member 2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so I can monitor academic operations.	4	High	Member 3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so I can quickly analyze system performance.	4	Medium	Member 4

Project Tracker, Velocity & Burndown Chart

Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase

Project Flow / Milestones

- Milestone 1: Salesforce Developer Account Creation
- Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
- Milestone 3: Tabs Creation
- Milestone 4: Lightning App Creation

- Milestone 5: Fields Creation
- Milestone 6: Create Relationships
- Milestone 7: Page Layout Customization
- Milestone 8: Validation Rules
- Milestone 9: Flows (Automation)
- Milestone 10: Approval Process
- Milestone 11: Reports
- Milestone 12: Dashboards
- Milestone 13: Security Setup
- Milestone 14: Conclusion

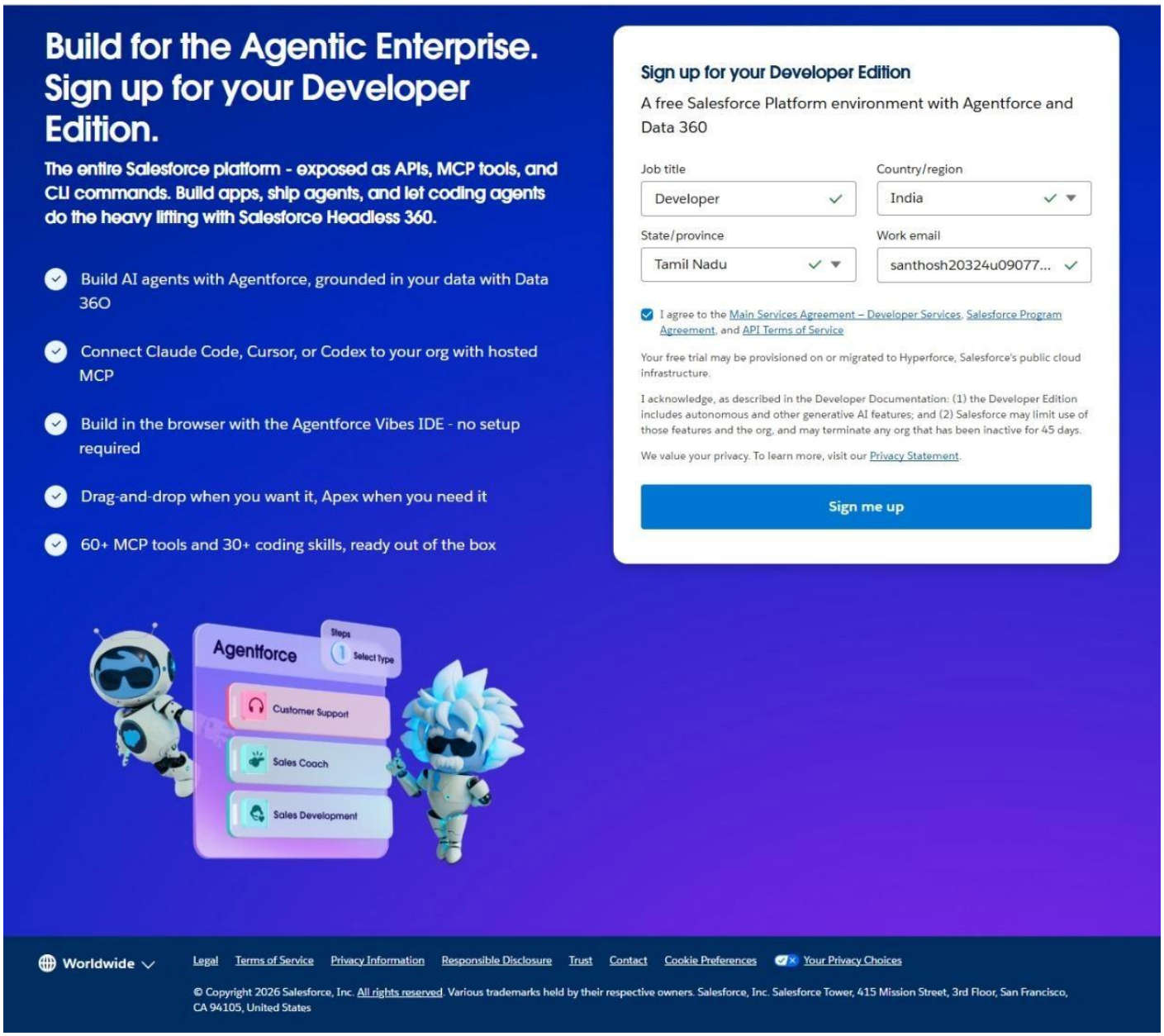
What You'll Learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships
- Creating page layouts & record types
- Writing validation rules
- Building automation with flows
- Creating approval workflows
- Developing reports & dashboards
- Setting up security with profiles, permission sets, and sharing
- End-to-end application testing

Activity 1: Creating a Developer Account

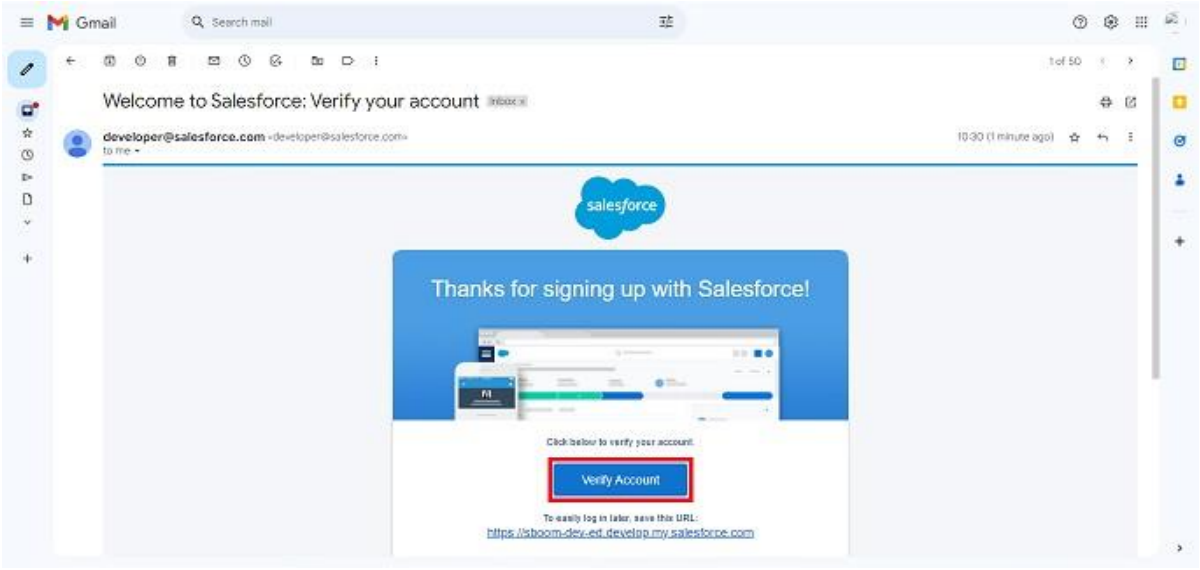
Creating a developer org in salesforce Go to
https://developer.salesforce.com/signup

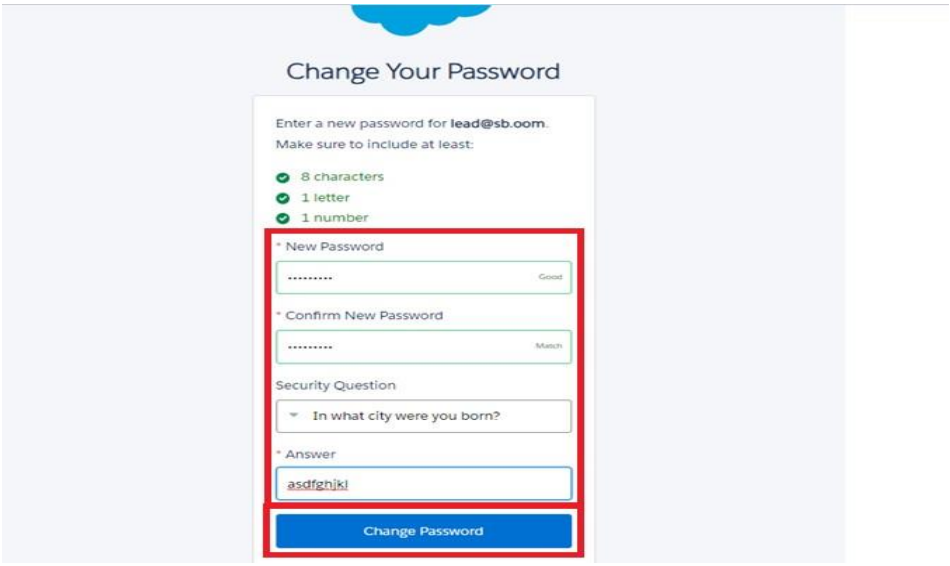
On the sign up form, enter the following details :



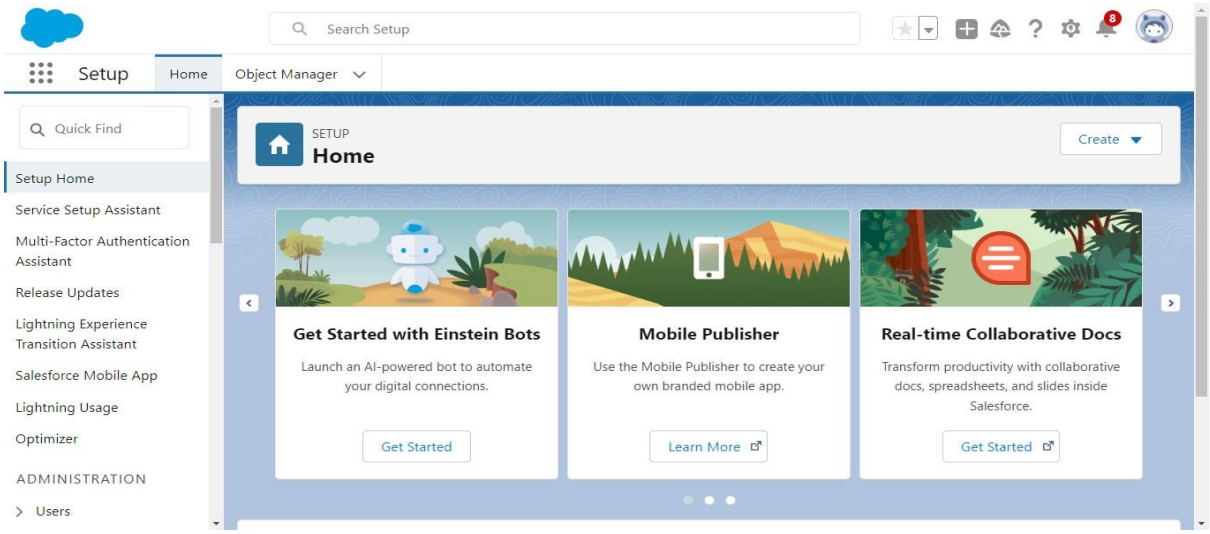
Activity 2: Account Activation

- Open the verification email and click **Verify Account**
- Set a password and answer a security question





Then you will redirect to your salesforce setup page.

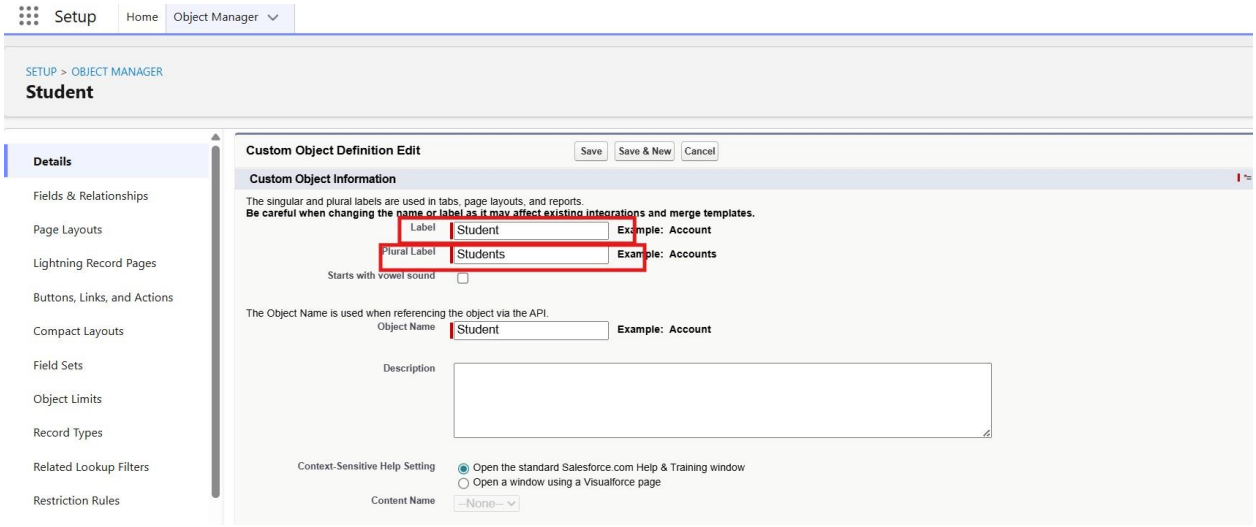


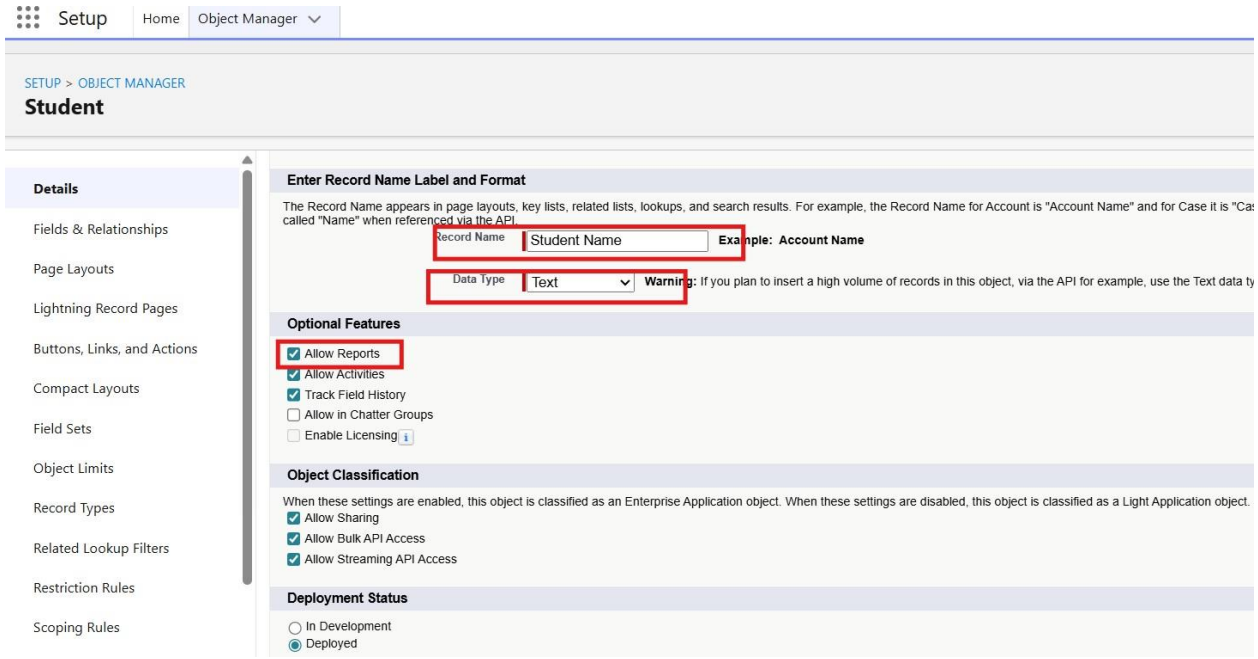
You are redirected to the Salesforce Setup page

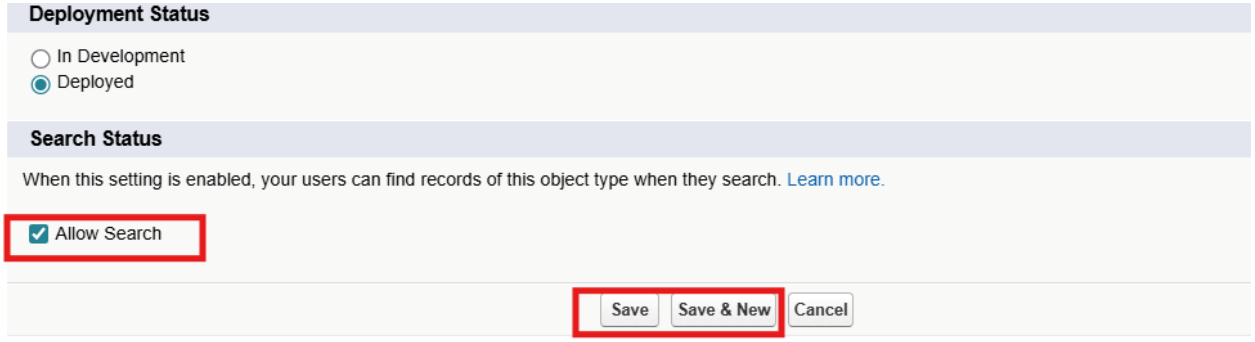
Activity 2: Creating the Student Object

Purpose: store and manage student personal details, contact information, and enrollment status.

- Go to Setup → Object Manager → Create → Custom Object
- Label: **Student**, Plural Label: **Students**, Object Name: **Student**
- Record Name: Student Name (Data Type: Text)
- Optional Features: Allow Reports, Allow Activities, Track Field History
- Search Status: Allow Search
- Click Save







The same process is repeated to create the **Course**, **Instructor**, and **Enrollment** custom objects.

Student Object Fields

Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Course Object Fields

Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Instructor Object Fields

Instructor Name	Text (Standard)
Instructor Code	Text

Expertise	Picklist
Phone	Phone
Email	Email

Enrollment Object Fields

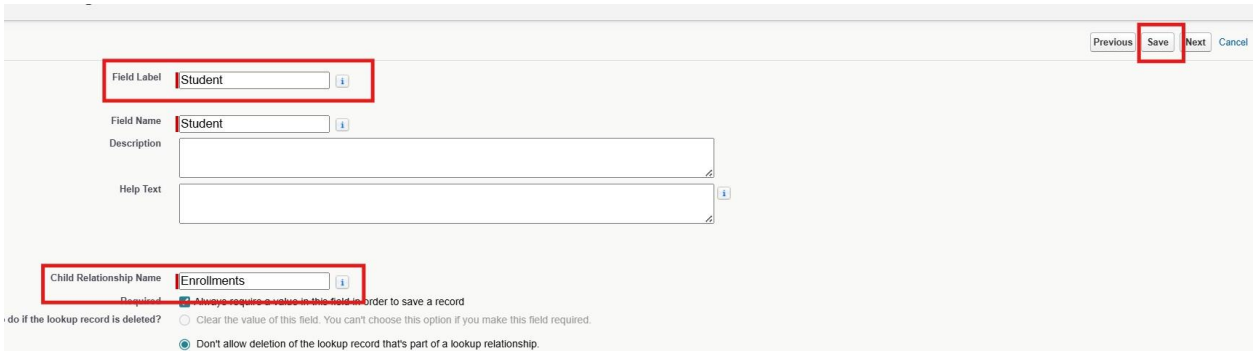
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Activity 1: Lookup from Enrollment to Student

- Setup → Object Manager → Enrollment__c → Fields & Relationships → New
- Select **Lookup Relationship** → Related To: Student__c
- Field Label: **Student** (API Name: Student__c)
- Configure field-level security and page layout visibilitySave

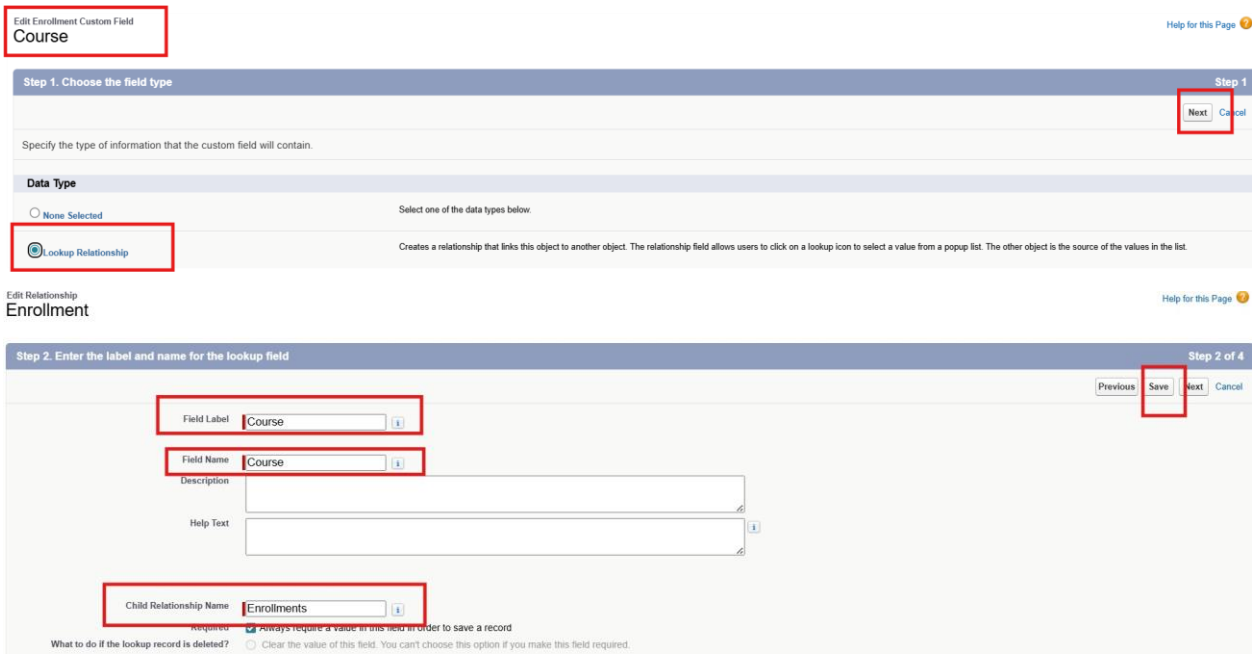


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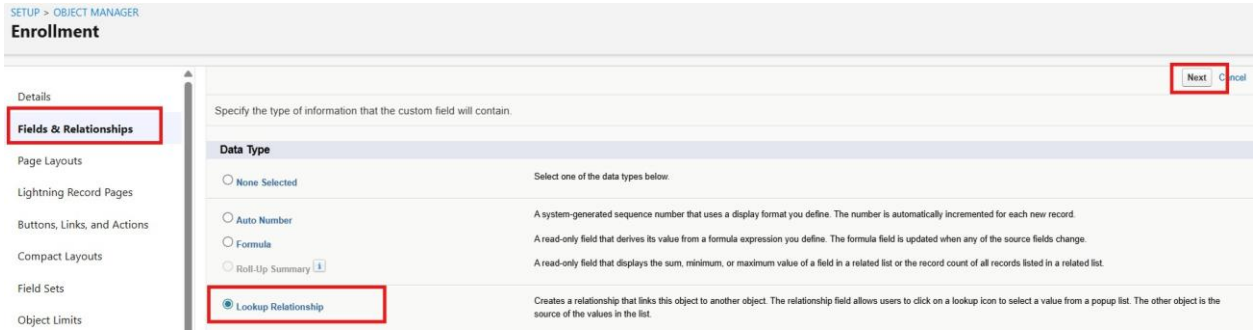
Activity 2: Lookup from Enrollment to Course

- Same steps, Related To: Course__c
- Field Label: **Course** (API Name: Course__c) → Save



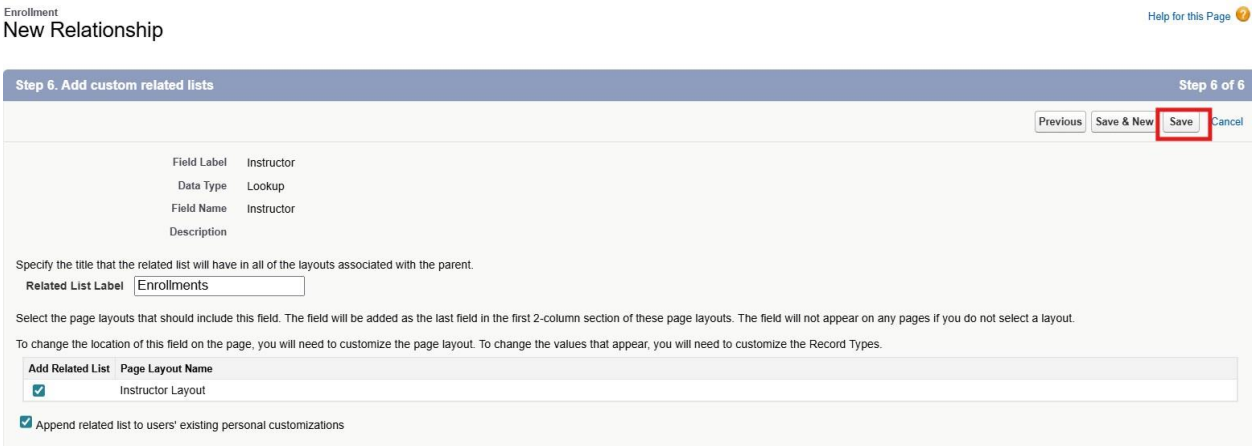
Activity 3: Lookup from Enrollment to Instructor

- Same steps, Related To: Instructor__c
- Field Label: **Instructor** (API Name: Instructor__c) → S







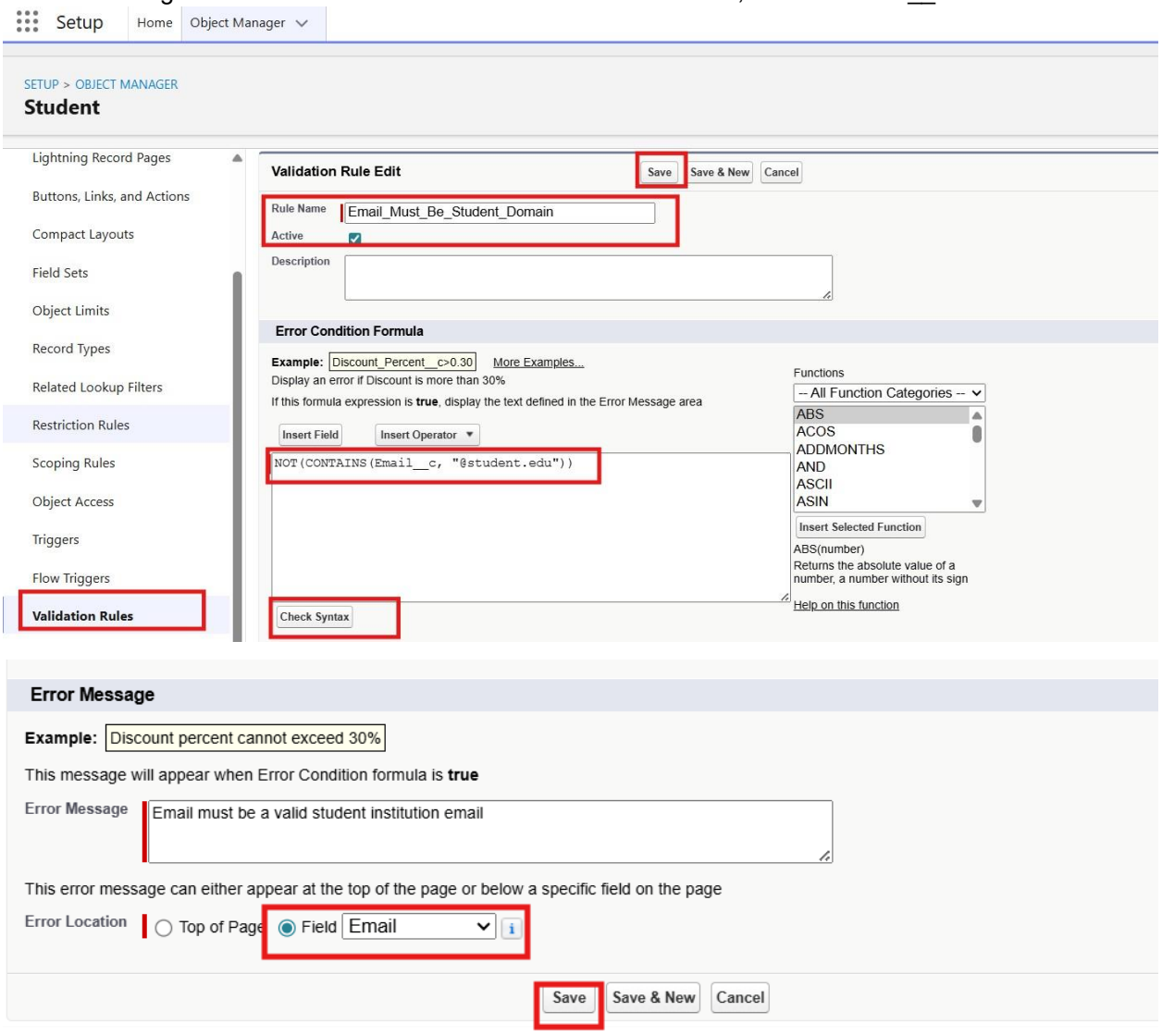


Activity 1: Student Email Domain Validation

Purpose: ensure students use only allowed institutional email addresses.

- Setup → Object Manager → Student__c → Validation Rules → New

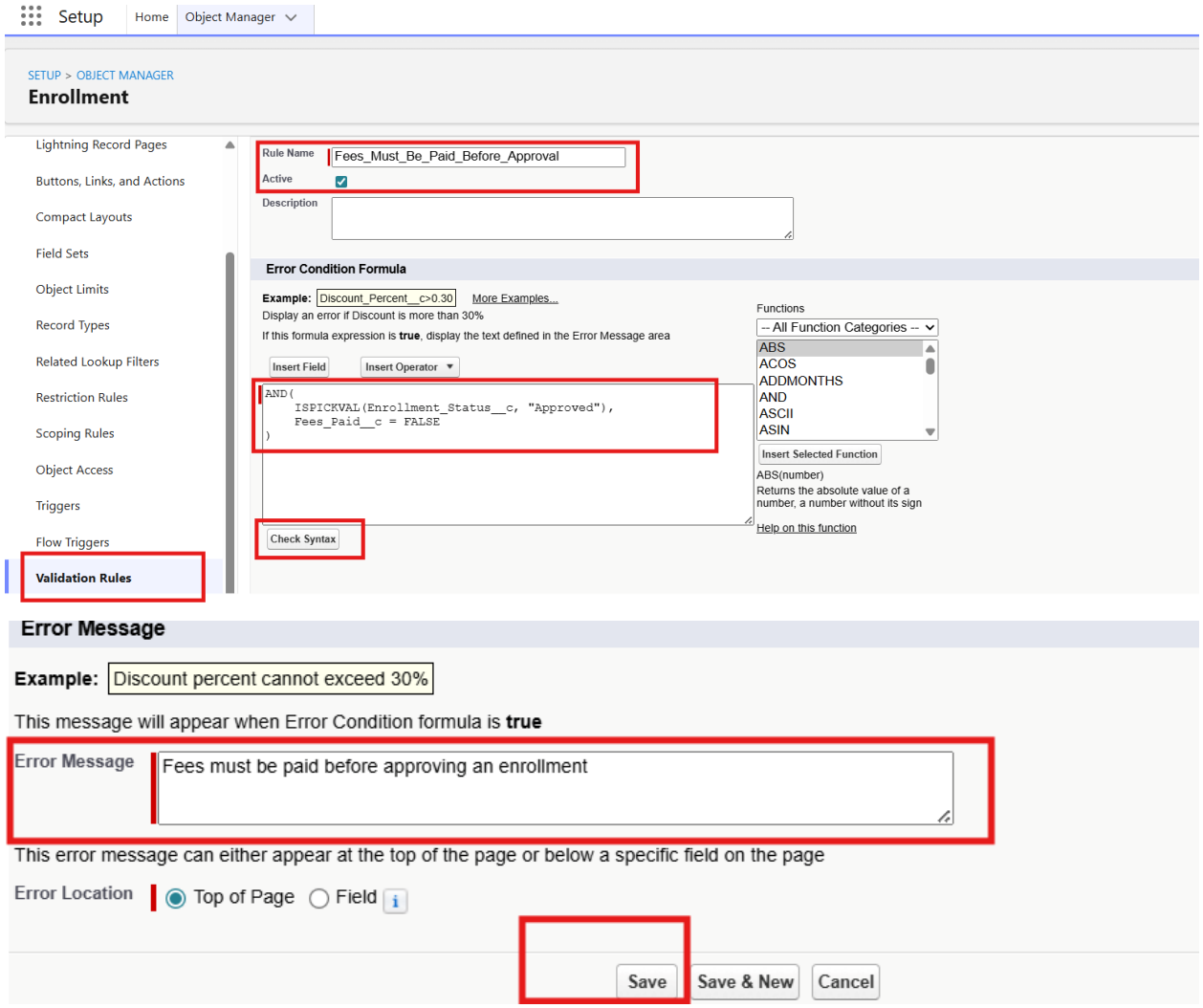
- Rule Name: **Student_Email_Domain_Validation**
- Formula: `NOT(CONTAINS(Email__c, "@student.edu"))`
- Error Message: “Email must be a valid student institutional email”, at Field: Email__c



Activity 2: Fees Must Be Paid Before Approval

Purpose: prevent an Enrollment from being approved until fees are paid.

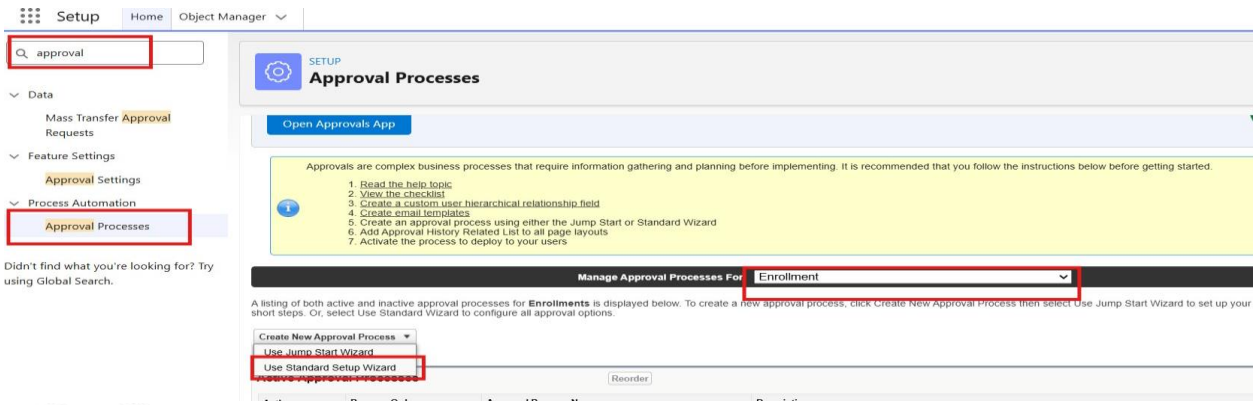
Rule Name: **Fees_Must_Be_Paid_Before_Approval**
Formula: `AND(ISPICKVAL(Enrollment_Status__c,"Approved"), Fees_Paid__c = FALSE)`
Error Message: “Fees must be paid before approving an enrollment”, at Top of Page



Activity 1: Approval Process for Enrollment Requests

Purpose: route Enrollment requests to the Training Manager for approval or rejection.

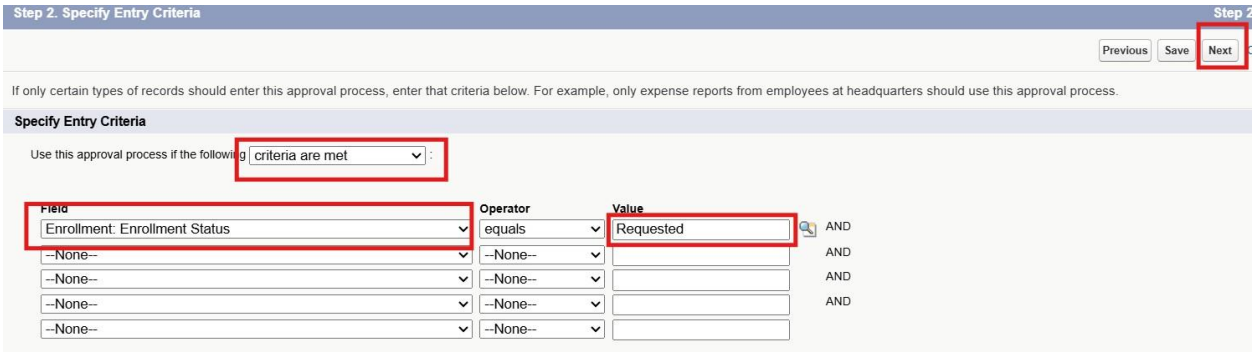
- Setup → Approval Processes → Enrollment__c → Create New (Standard Setup)
- Process Name: **Approval_Enrollment_Request**
- Entry Criteria: Enrollment_Status__c = “Requested”
- Approval Step: assign a specific Training Manager as approver



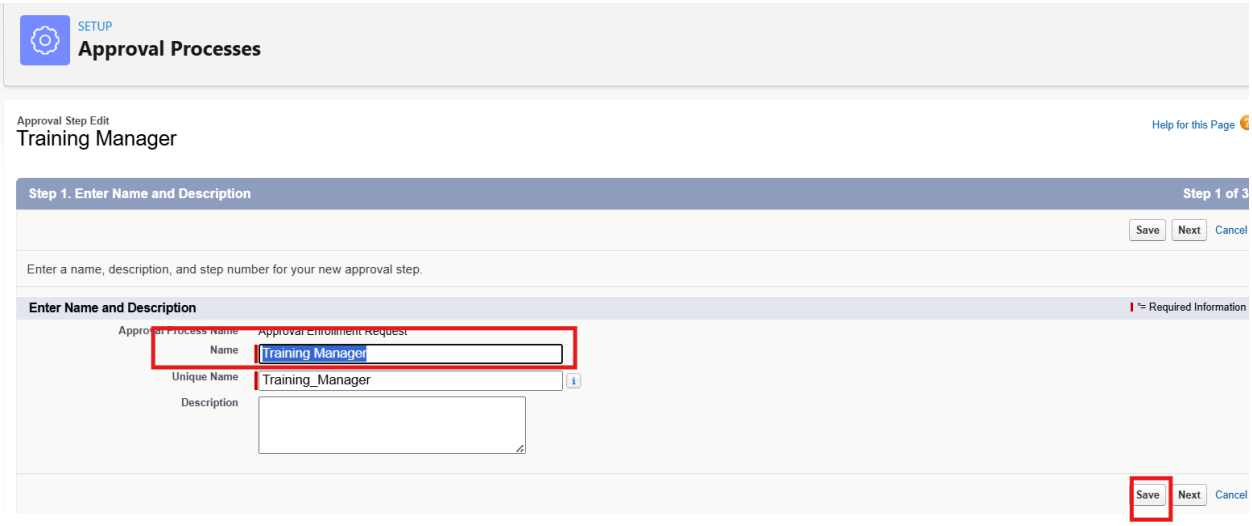
Approval Process Edit

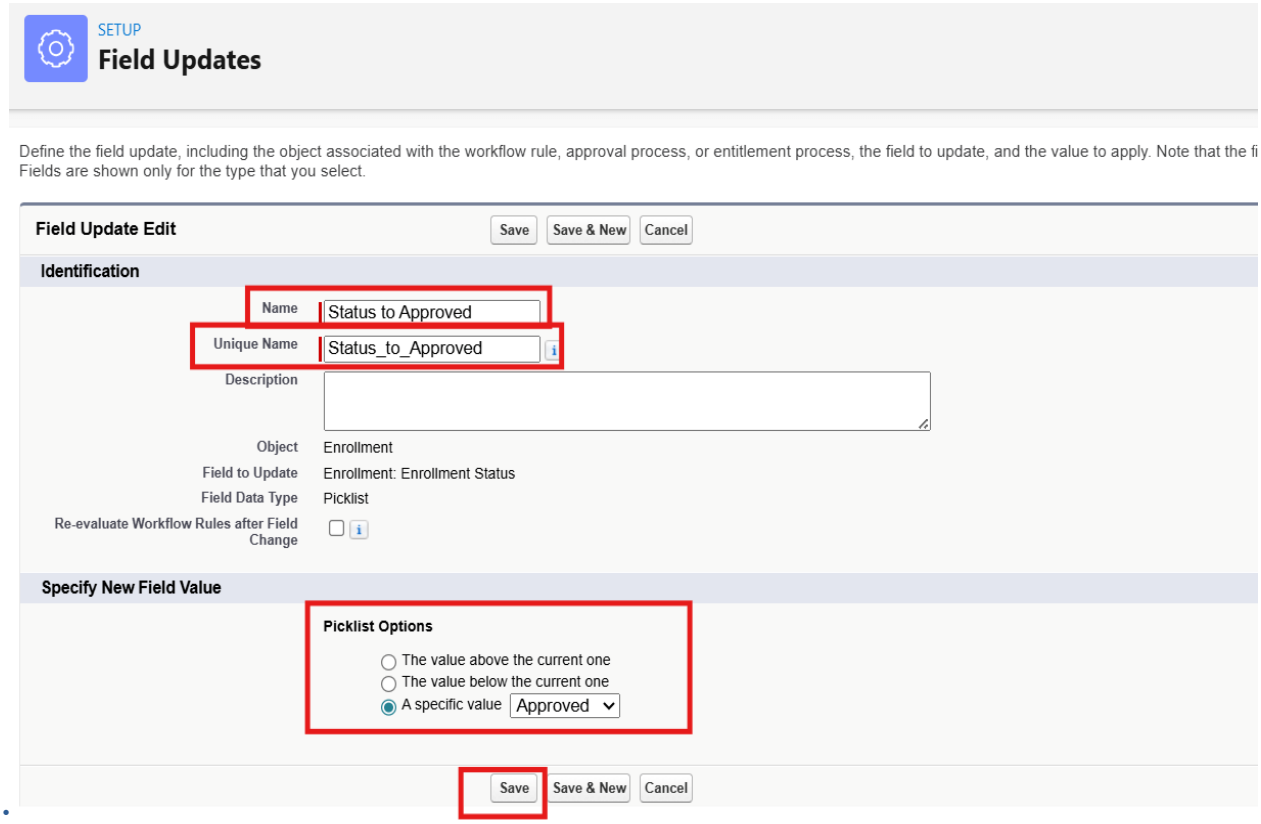
Approval Enrollment Request



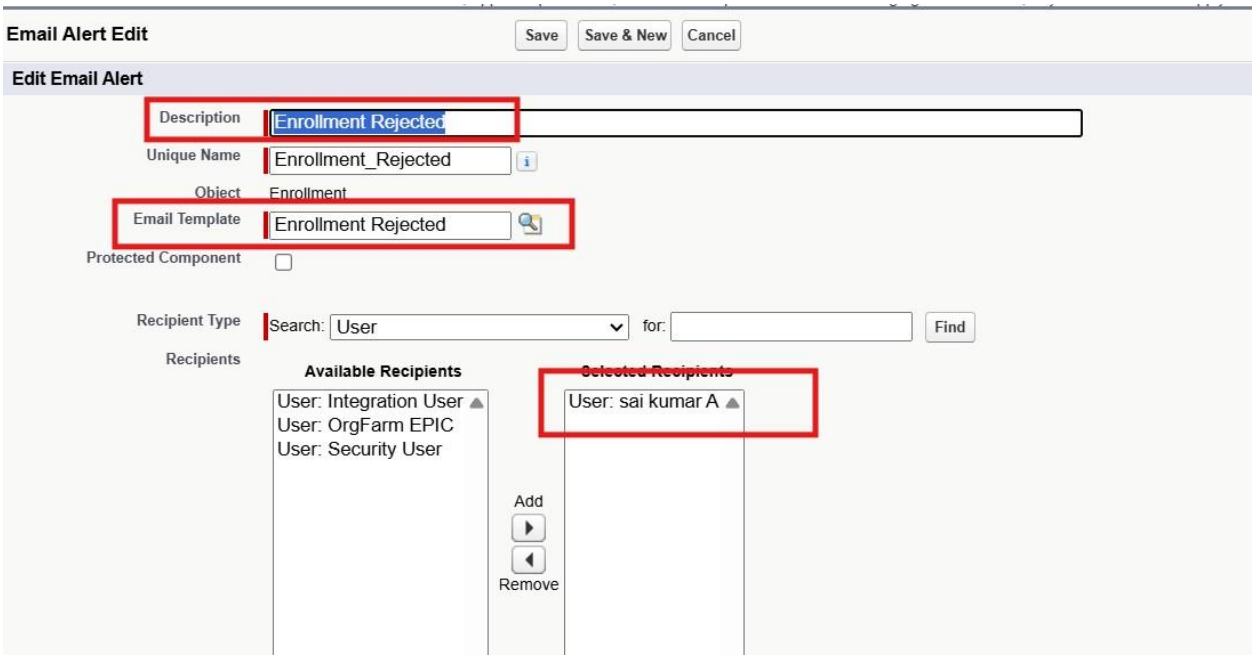


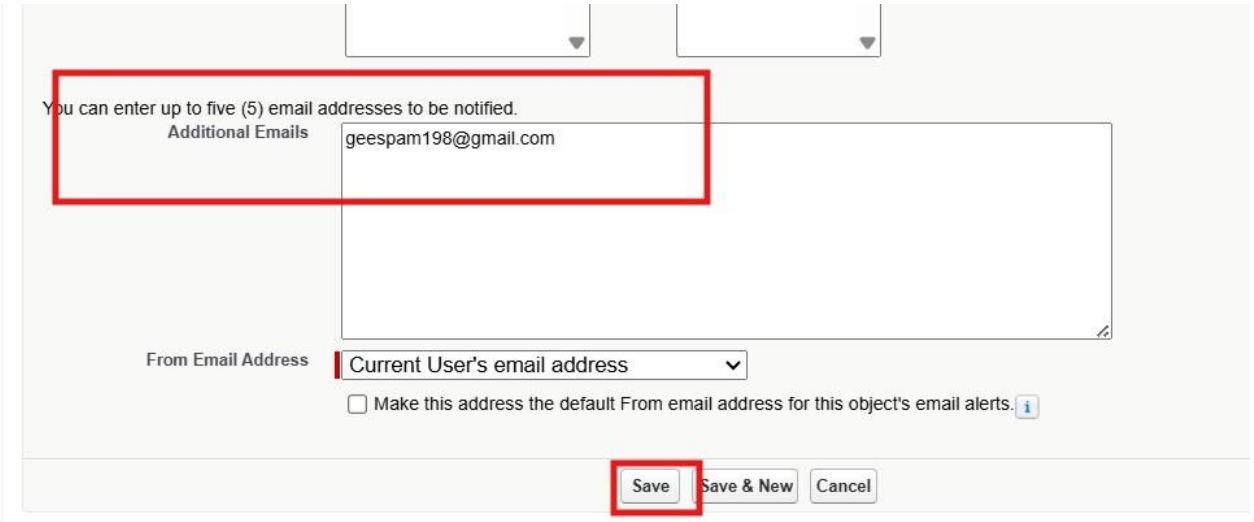
- Final Approval Action: Field Update — Enrollment_Status__c = “Approved”

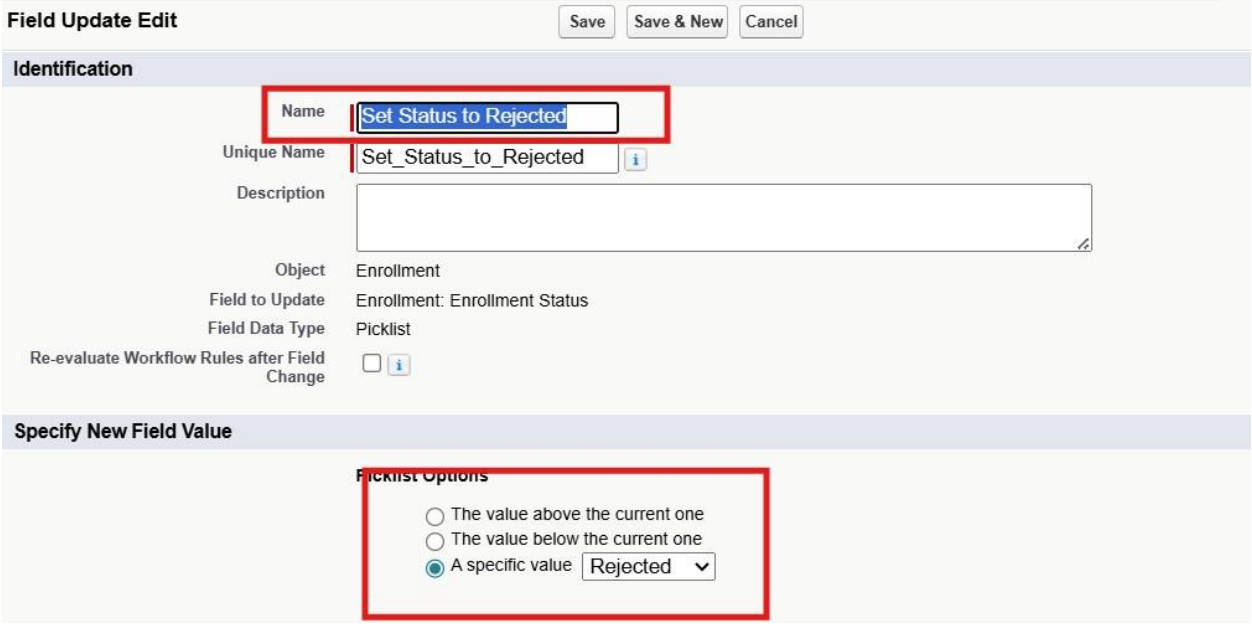




- Final Rejection Action: Field Update — Enrollment_Status__c = “Rejected”, plus an Email Alert to the student



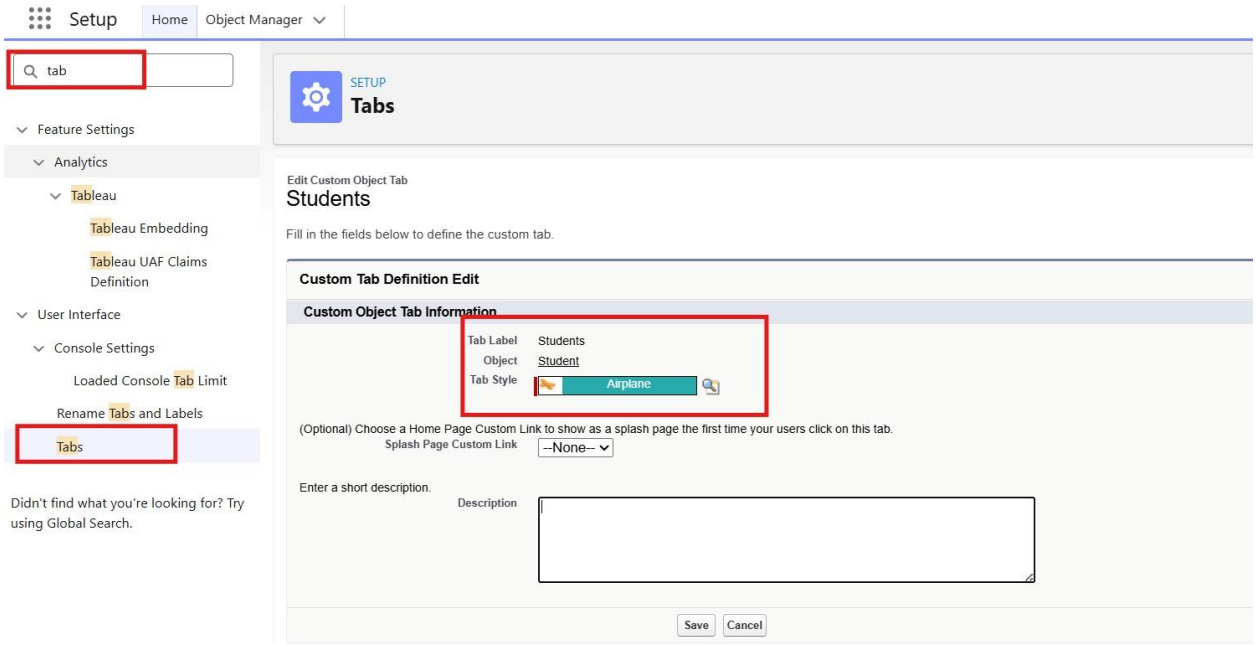




Activate the approval process

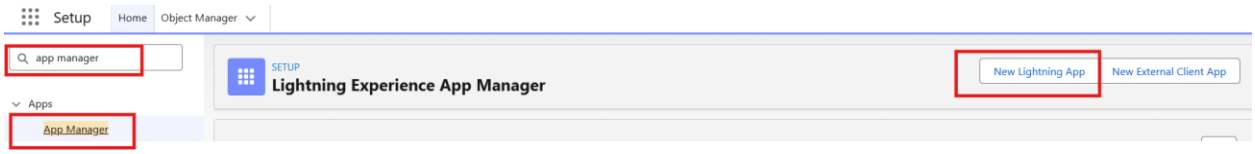
Activity 1 & 2: Creating Tabs

Setup → Tabs → Custom Object Tabs → New
Select each object (Student, Course, Instructor, Enrollment), choose a tab style/icon
Assign to all profiles and Save



Activity 1: Creating the Lightning Application

Setup → App Manager → New Lightning App
App Name: **Student Enrollment & Course Management**



App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name
Student Enrollment & Course Management

* Developer Name
Student_Enrollment_Course_Management

Description
Enter a description...

App Branding

Image



Clear

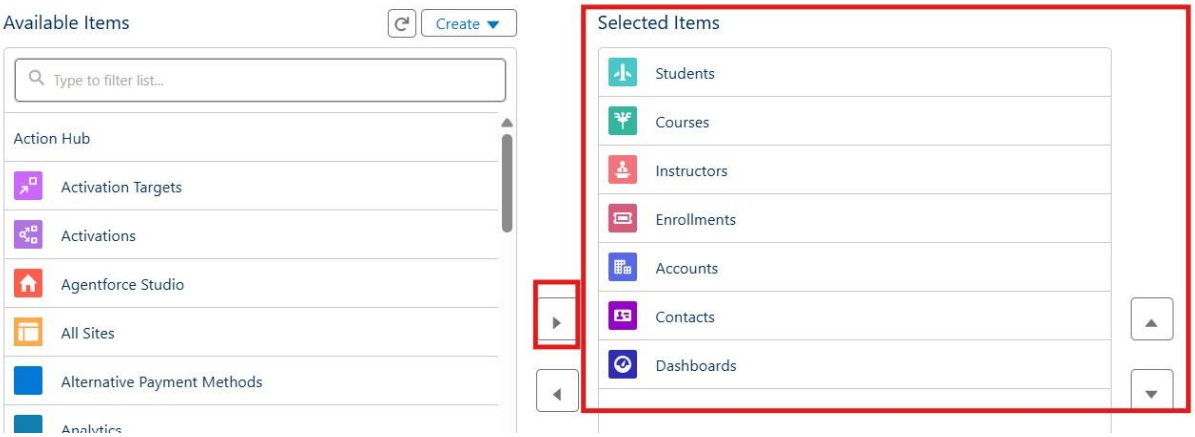
Primary Color Hex Value
#0070D2

Org Theme Options

Use the app's image and color instead of the org's custom theme

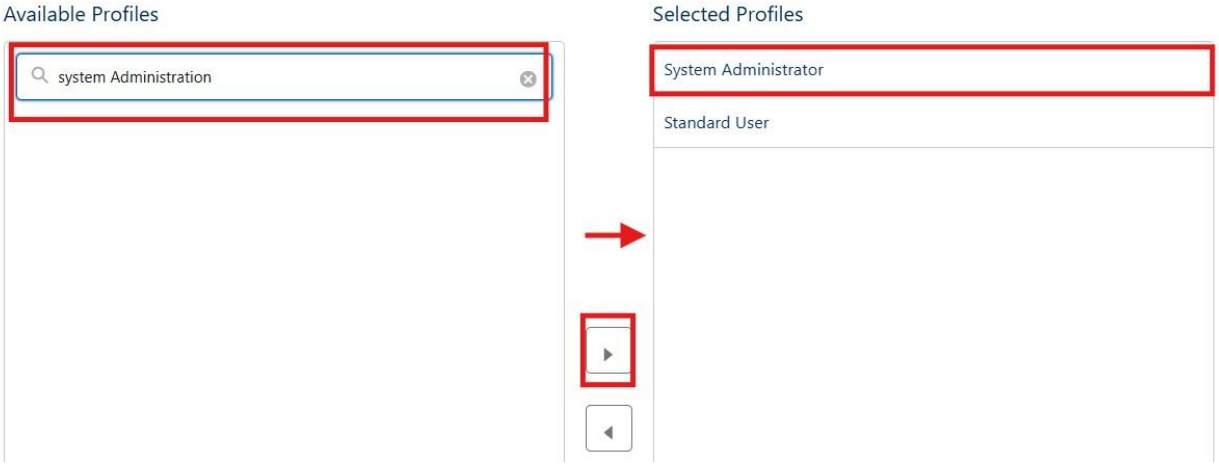
Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.



User Profiles

Choose the user profiles that can access this app.



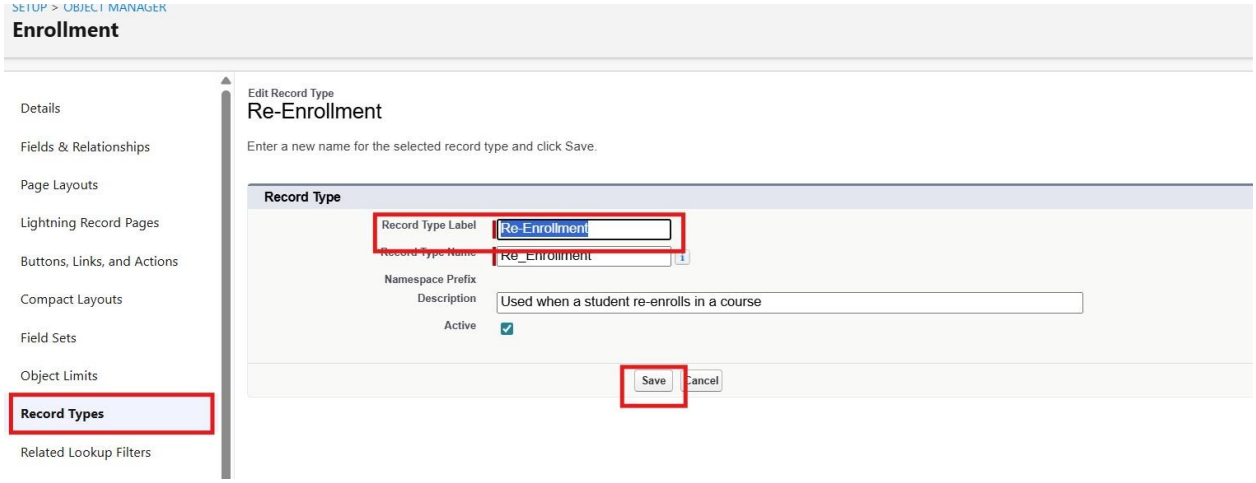
- Navigation Style: Standard Navigation
- Assign to profiles: Admin, Enrollment Officer, Instructor
- Navigation Items: Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, Tasks

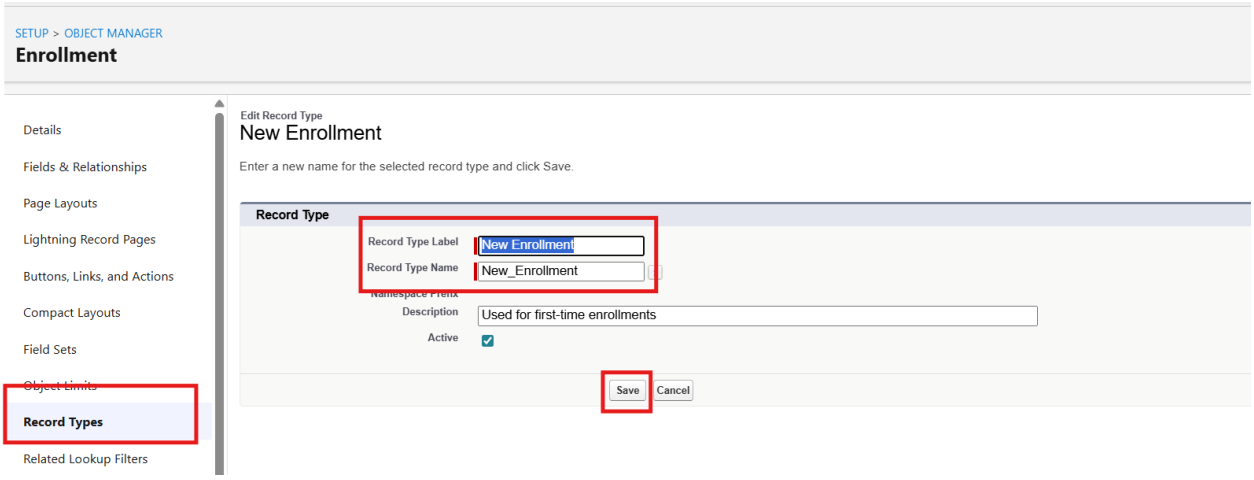
Review settings → Save & Finish

Activity 1: Record Types for Enrollment

Purpose: separate New Enrollment vs Re-Enrollment processes.

Setup → Object Manager → Enrollment__c → Record Types → New
Record Type Label: **New Enrollment** (from Master) — enable for Admin, Enrollment Officer
Record Type Label: **Re-Enrollment** — enable for the same profiles

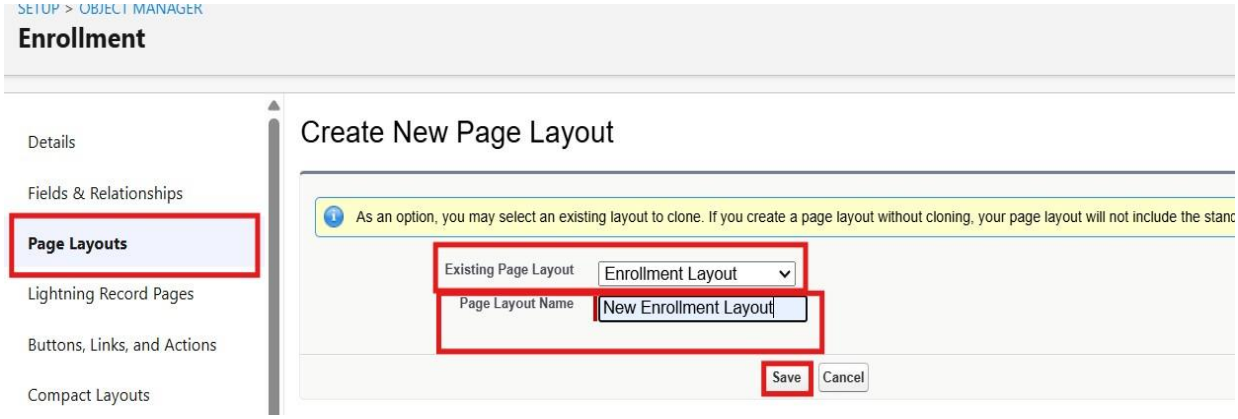


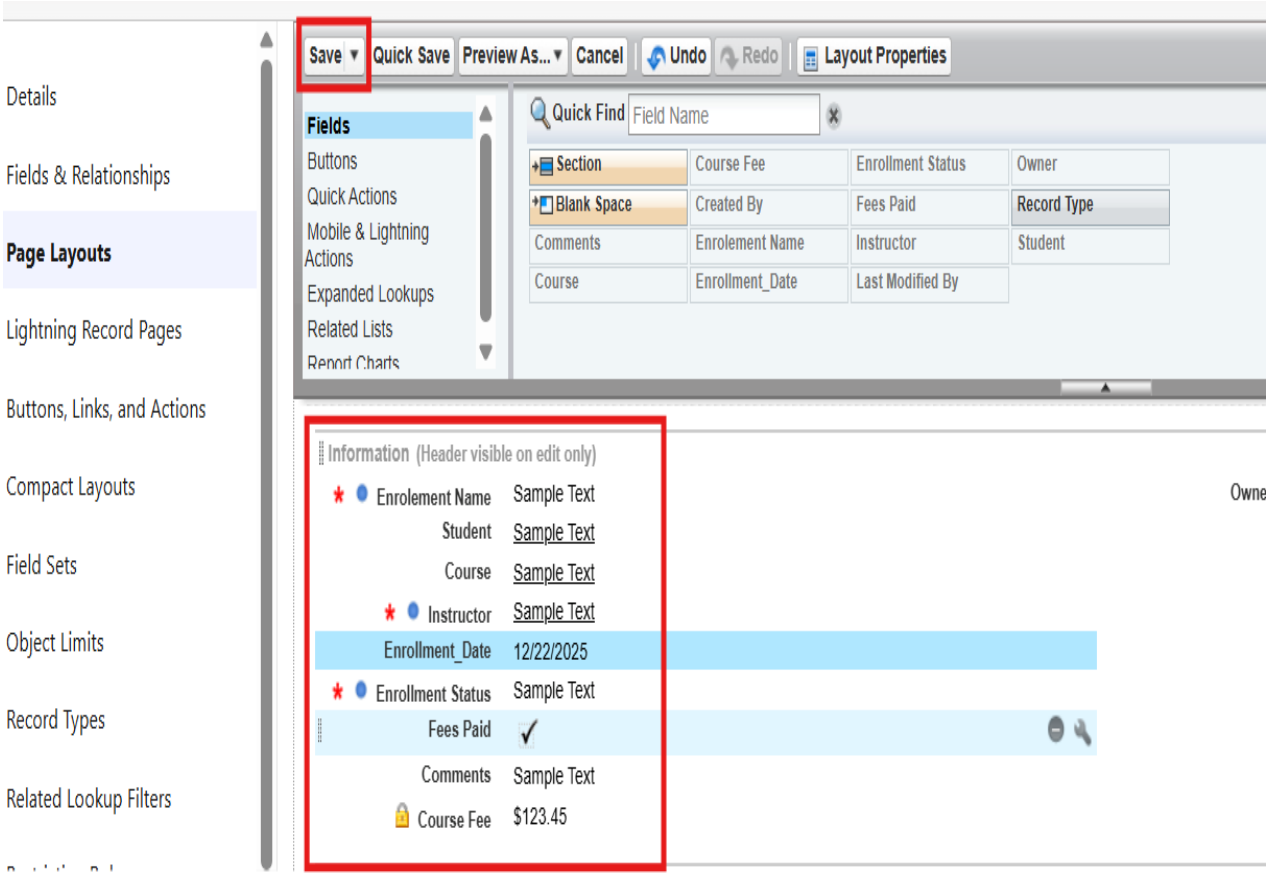


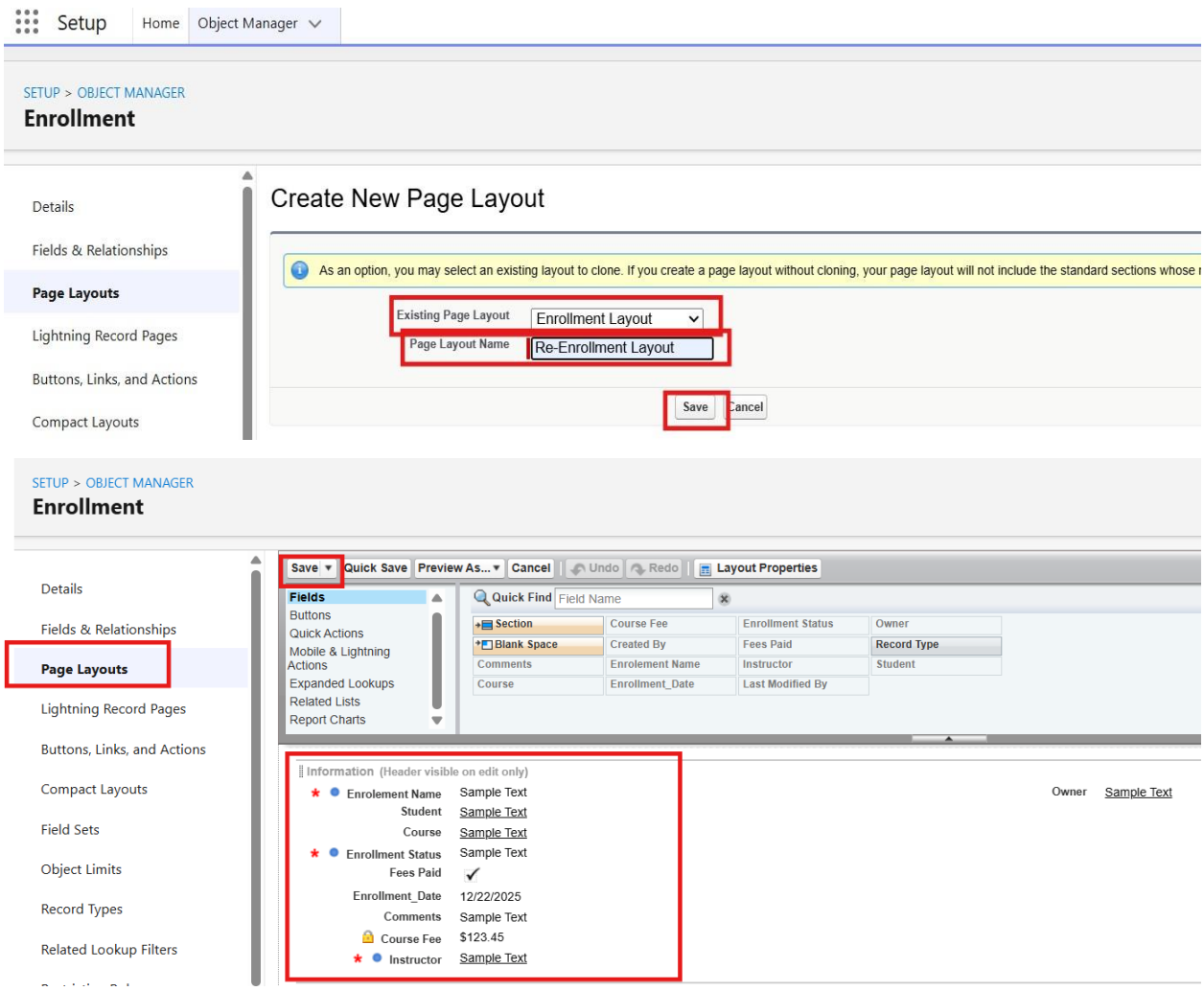
Activity 2: Separate Page Layouts per Record Type

- Clone the Enrollment Layout → name it **New Enrollment Layout**; keep fields relevant to first-time enrollment

(Student, Course, Instructor, Enrollment Date, Status, Fees)


SETUP > OBJECT MANAGER
Enrollment
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Create New Page Layout
As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the stanc
Existing Page Layout
Enrollment Layout
Page Layout Name
New Enrollment Layout
Save
Cancel


Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Save
Quick Save
Preview As...
Cancel
Undo
Redo
Layout Properties
Fields
Buttons
Quick Actions
Mobile & Lightning Actions
Expanded Lookups
Related Lists
Report Charts
Quick Find
Field Name
Section
Course Fee
Enrollment Status
Owner
Blank Space
Created By
Fees Paid
Record Type
Comments
Enrolement Name
Instructor
Student
Course
Enrollment_Date
Last Modified By
Information (Header visible on edit only)
Enrolement Name
Sample Text
Student
Sample Text
Course
Sample Text
Instructor
Sample Text
Enrollment_Date
12/22/2025
Enrollment Status
Sample Text
Fees Paid
Comments
Sample Text
Course Fee
$123.45
Owne



Clone again → name it **Re-Enrollment Layout**; include re-enrollment specific fields • Page Layout Assignment: map

New Enrollment → New Enrollment Layout, Re-Enrollment → Re-Enrollment Layout

Activity 3: Related Lists on Parent Objects

Purpose: view all Enrollments from Student, Course, and Instructor records.

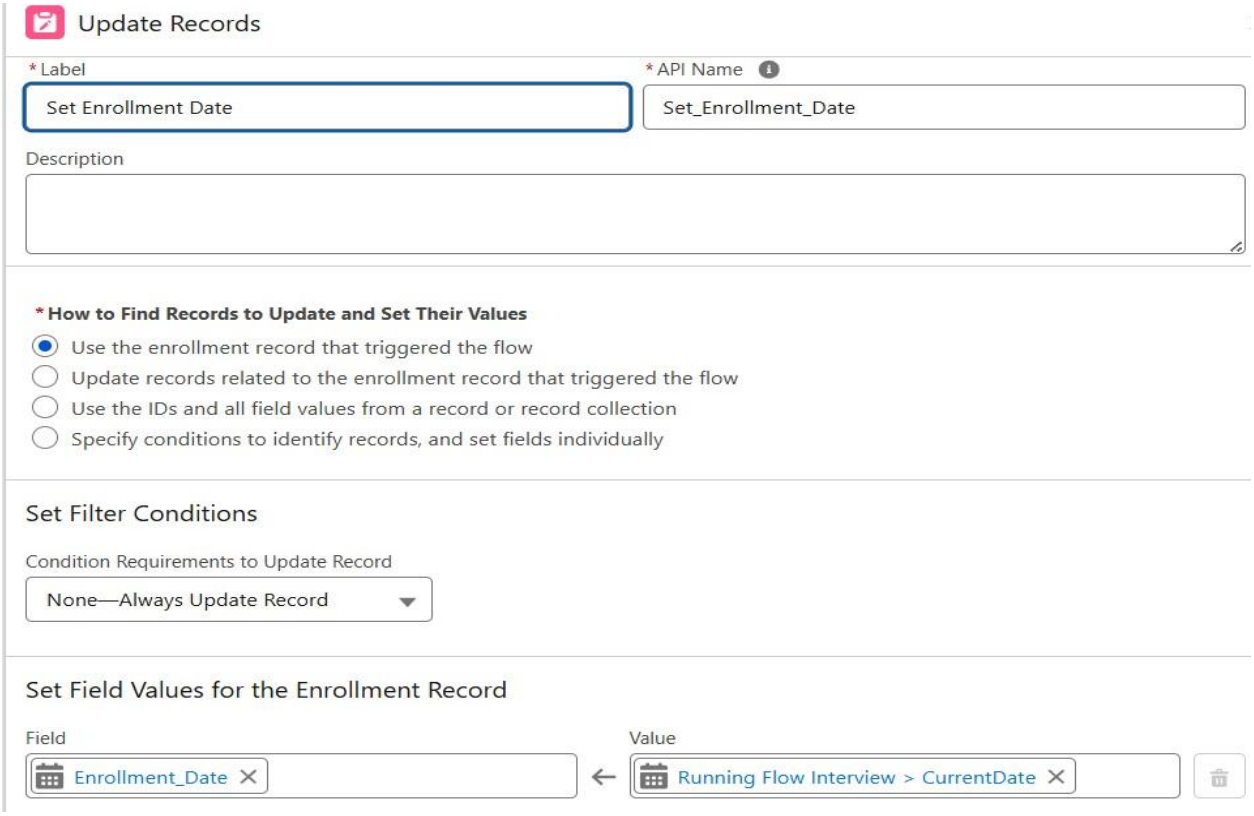
- Add the Enrollments related list to the Student__c page layout
- Add the Enrollments related list to the Course__c page layout
- Add the Enrollments related list to the Instructor__c page layout

Activity 1: Auto-Populate Enrollment Date

Purpose: set Enrollment_Date__c to today when an Enrollment record is created.

- Setup → Flows → New Flow → Record-Triggered Flow on Enrollment__c
- Trigger: A record is created
- Update Records: Enrollment_Date__c = $Flow.CurrentDate

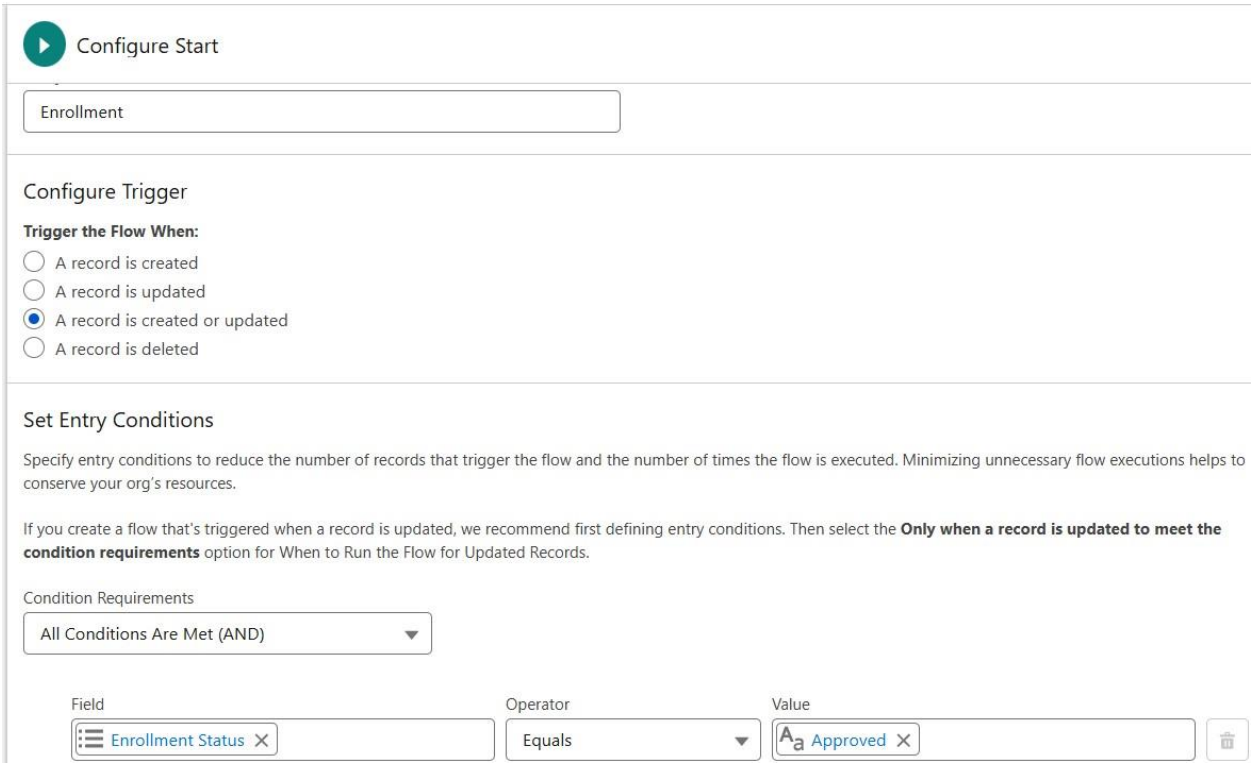
- Save as **SetEnrollmentDate** → Activate



Activity 2: Send Email When Enrollment Approved

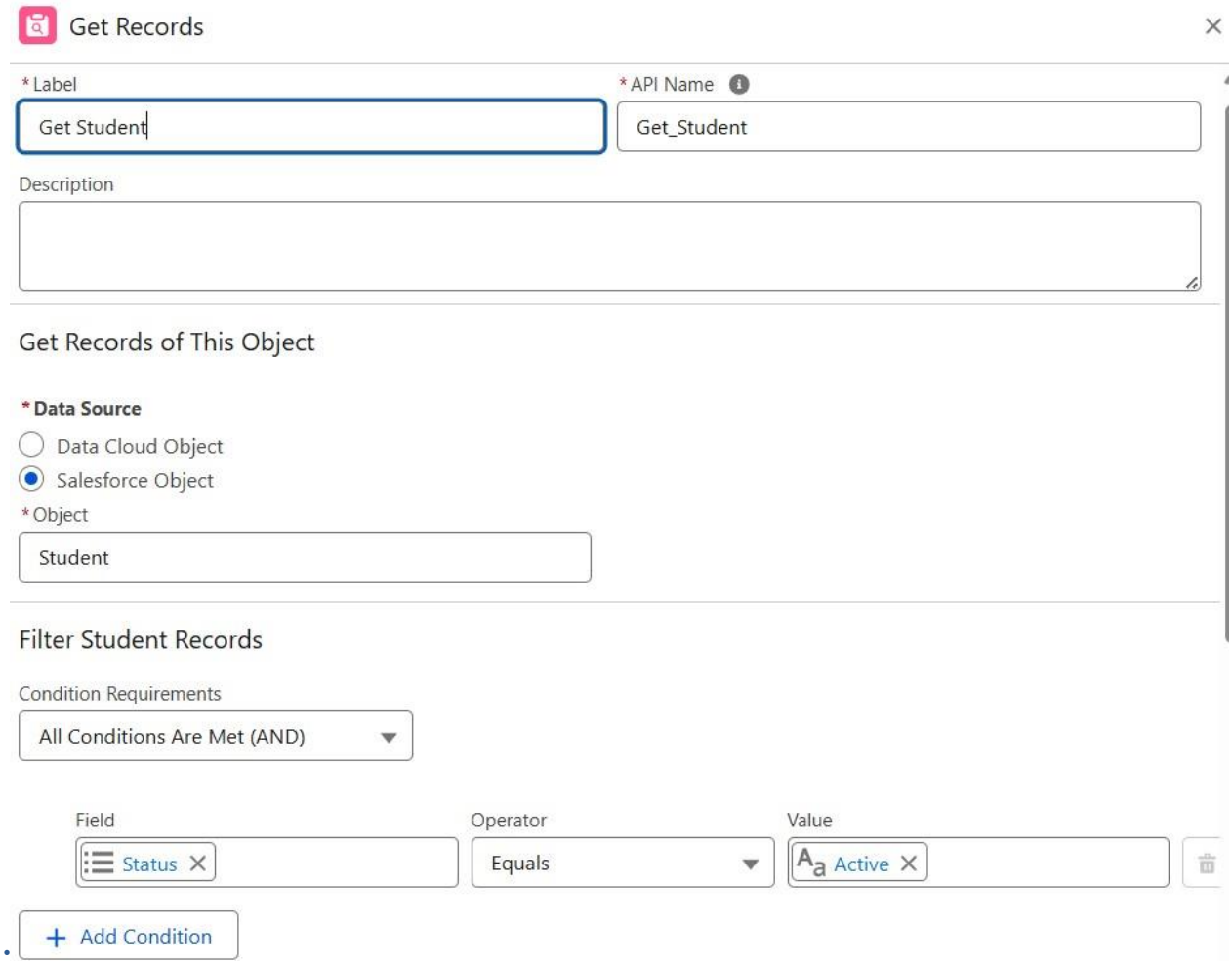
Purpose: when Enrollment_Status__c = Approved, send an email, alert admin, and set the Student to Active.

- Record-Triggered Flow on Enrollment__c, trigger on update, condition: Enrollment_Status__c Equals



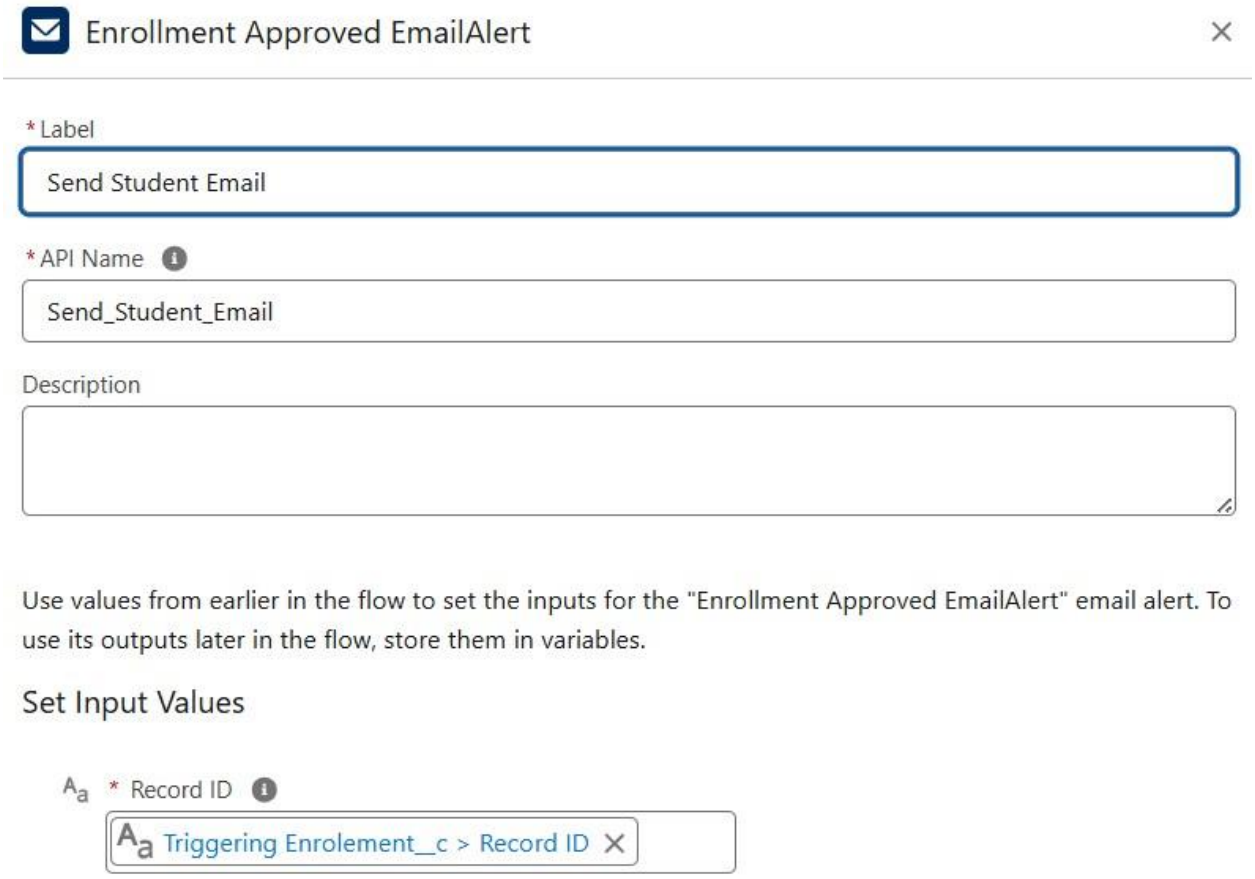
"Approved"
-

Get Records: fetch related Student__c

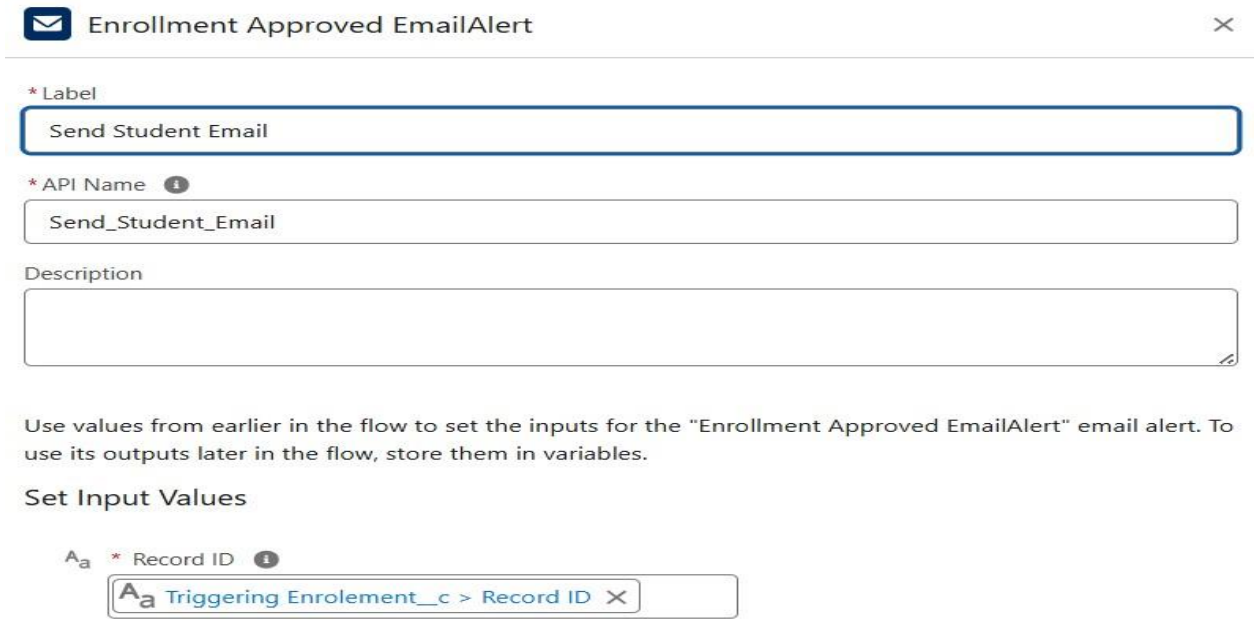


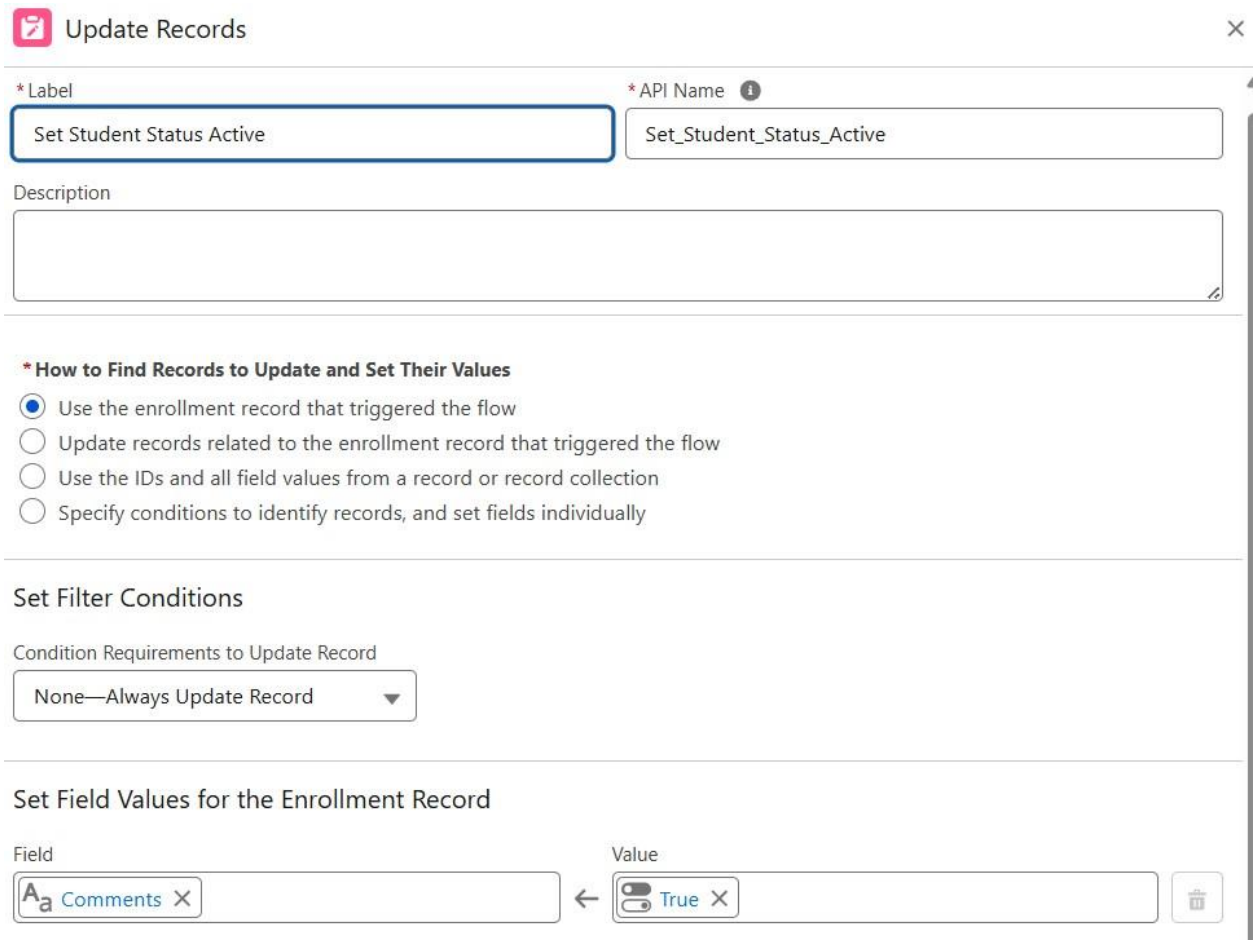
Action: send Email Alert to the student

Action: notify Admin if needed



Update Records: Student__c.Status__c = “Active”





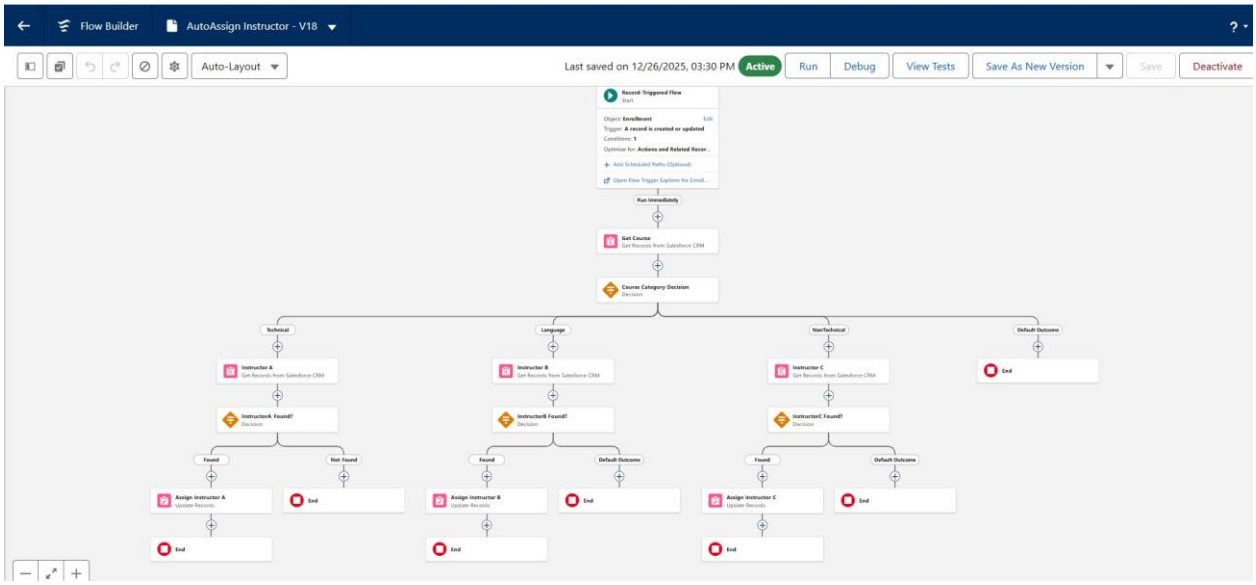
- Save as **Enrollment Approved Action** → Activate

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose: assign an instructor automatically based on Course category.

Technical	Instructor A
Language	Instructor B
Non-Technical	Instructor C

- Prepare Instructor__c records with codes (INSTR_A, INSTR_B, INSTR_C)
- Record-Triggered Flow on Enrollment__c (created/updated)
- Get Records: related Course__c; Decision on Course__r.Category__c
- For each outcome, Get the matching Instructor record and Update Records: Instructor__c = Get_Instructor_X.Id

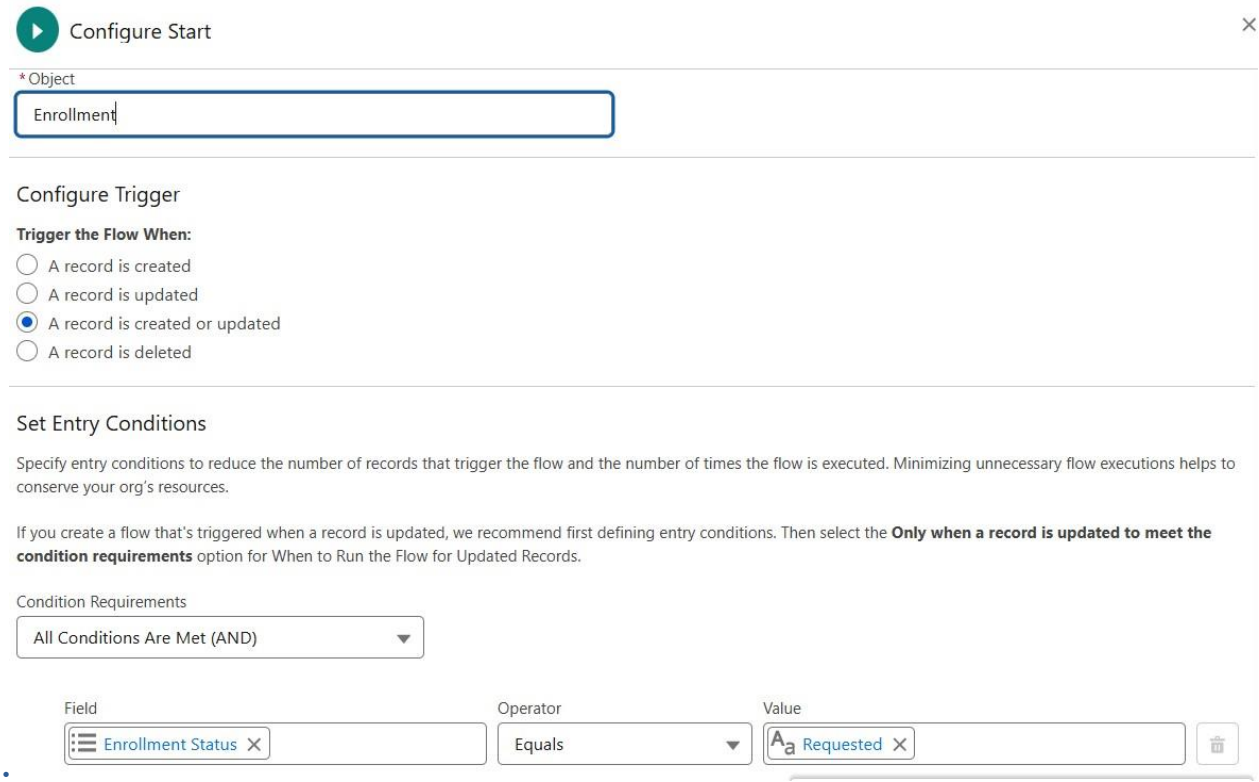


- Save as **AutoAssign_Instructor** → Activate

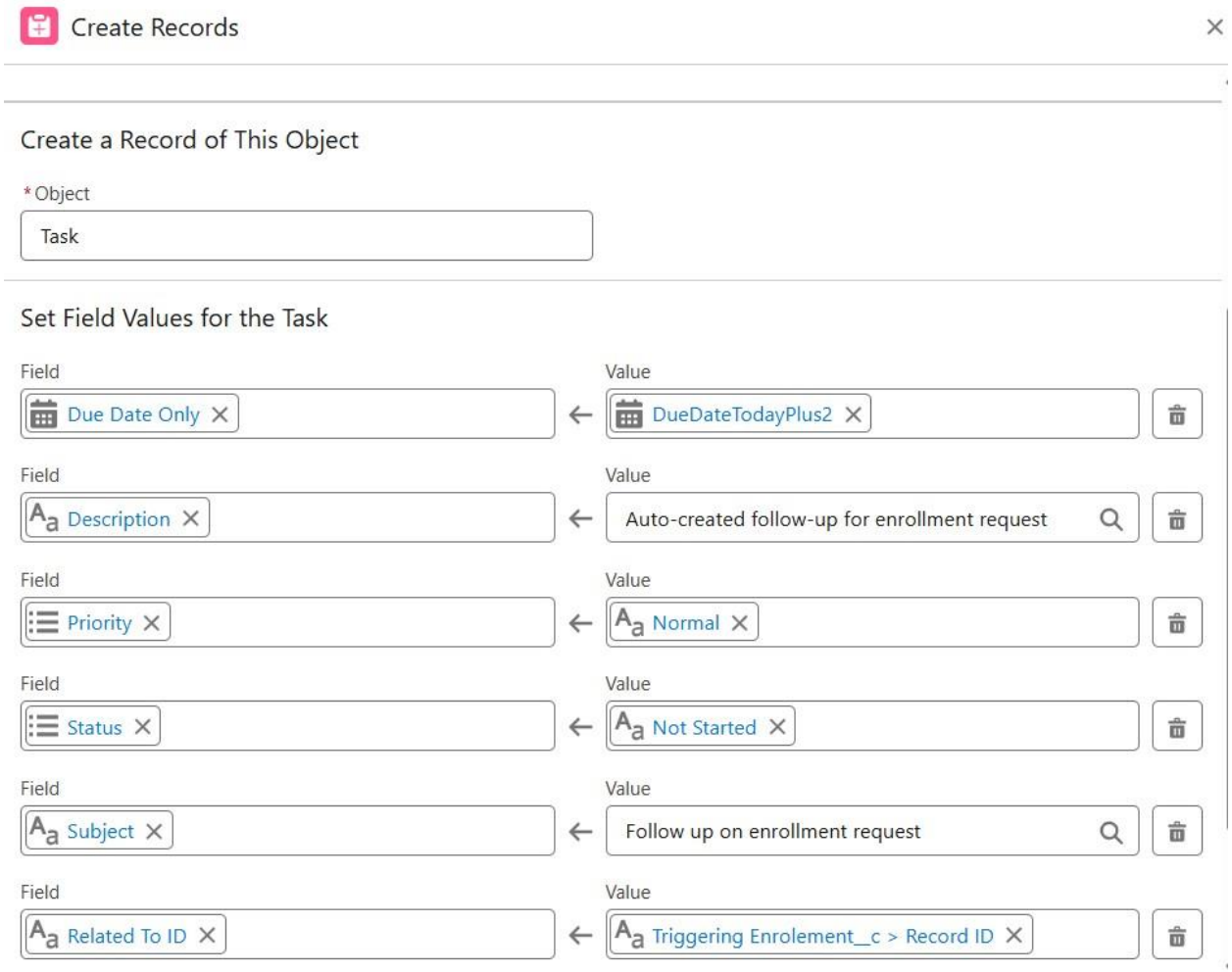
Activity 4: Auto-Create Task for Follow-Up

Purpose: when Enrollment is “Requested”, create a follow-up task for Admin.

- Record-Triggered Flow on Enrollment__c; condition: Enrollment_Status__c = “Requested” Get Records: Admin user (by Username/Profile)



- Formula (Date): TODAY() + 2 for the due date
- Create Records (Task): Subject “Follow up on enrollment request”, OwnerId = Admin user Id, WhatId = Enrollment Id, ActivityDate = Today + 2

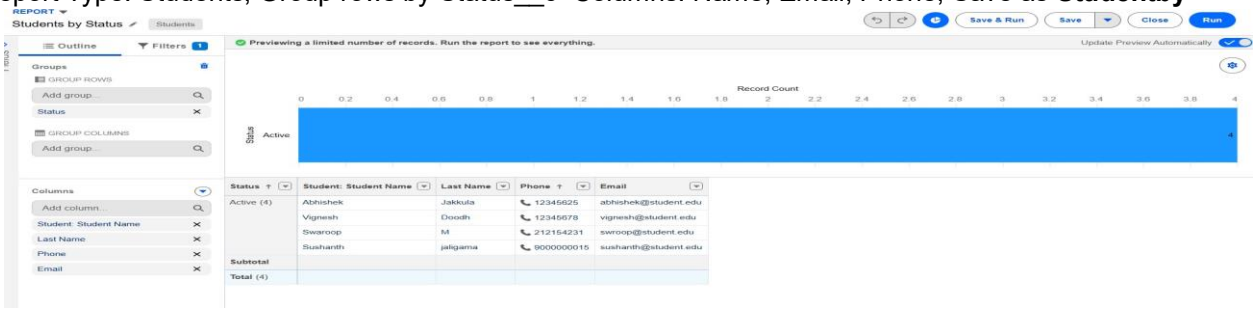


- → Activate

SECMS Documentation Salesforce Project Save as **Create_Followup_Task**

Activity 1: Students by Status

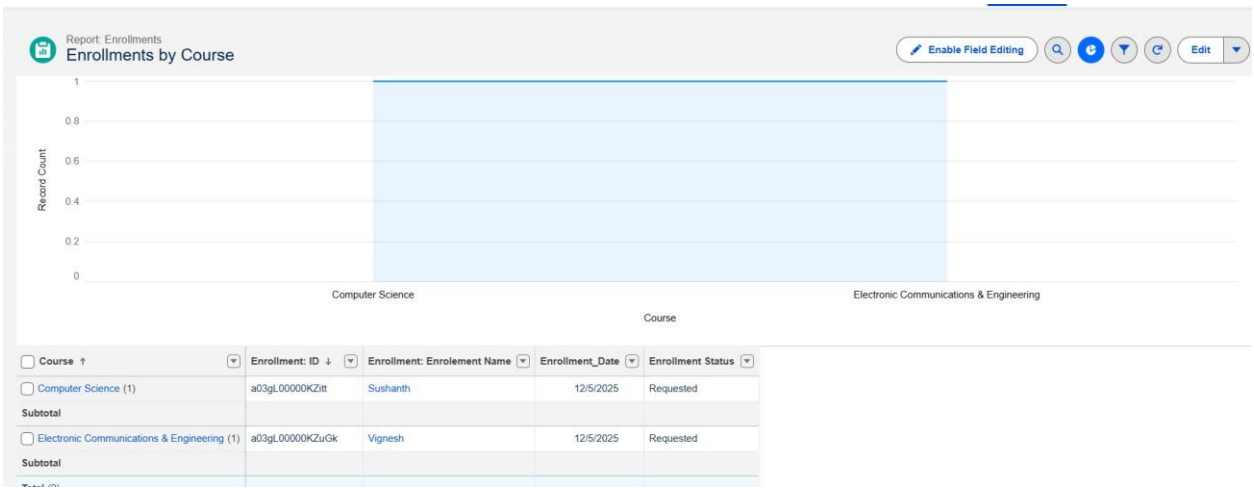
- Report Type: Students; Group rows by Status__c Columns: Name, Email, Phone; Save as **Studentby**



- Columns: Name, Email, Phone; Save as **Studentsby S tatus**

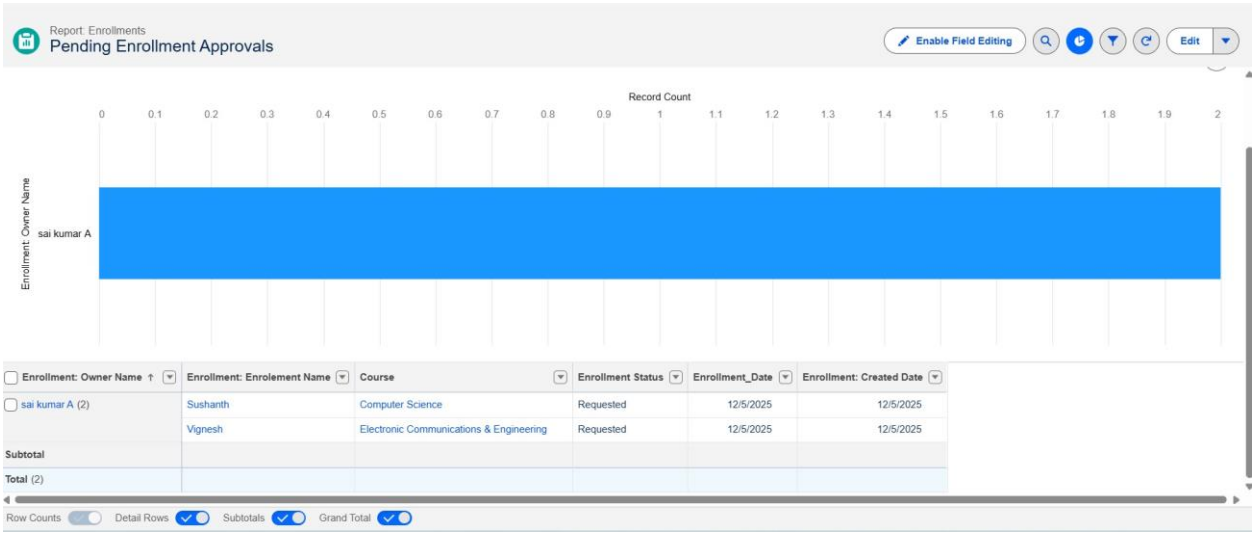
Activity 2: Enrollments by Course

- Report Type: Enrollments; Group rows by Course__r.Name
- Show record count per course; Save as **Enrollments by Course**



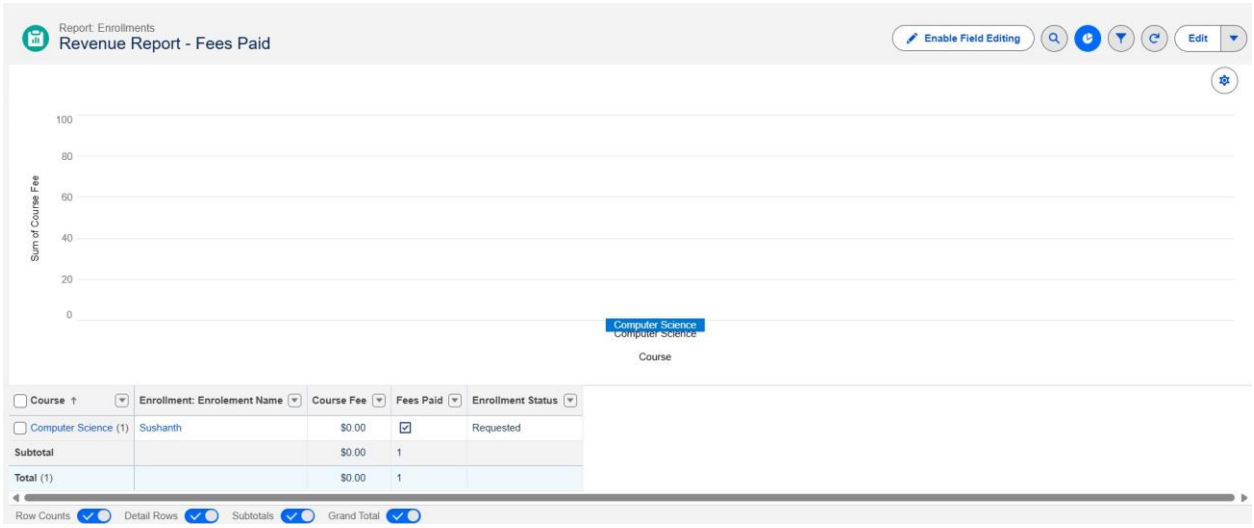
Activity 3: Pending Enrollment Approvals

- Report Type: Enrollments; Filter: Enrollment_Status__c = “Requested”
- Columns: Student, Course, Owner, Created Date; Save as **Pending Enrollment Approvals**



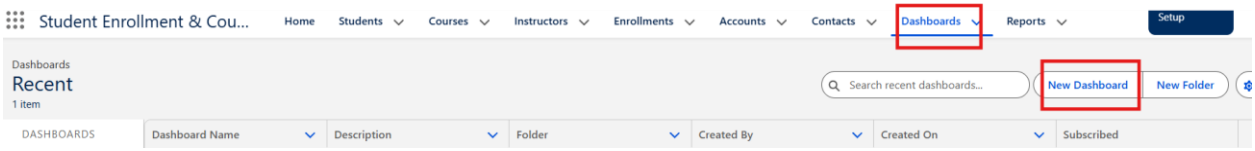
Activity 4: Revenue Report (Fees Paid)

- Report Type: Enrollments; Filter: Fees_Paid__c = TRUE
- Group by Course__r.Name; summarize Course_Fee__c
- Save as **Revenue Report – Fees Paid**

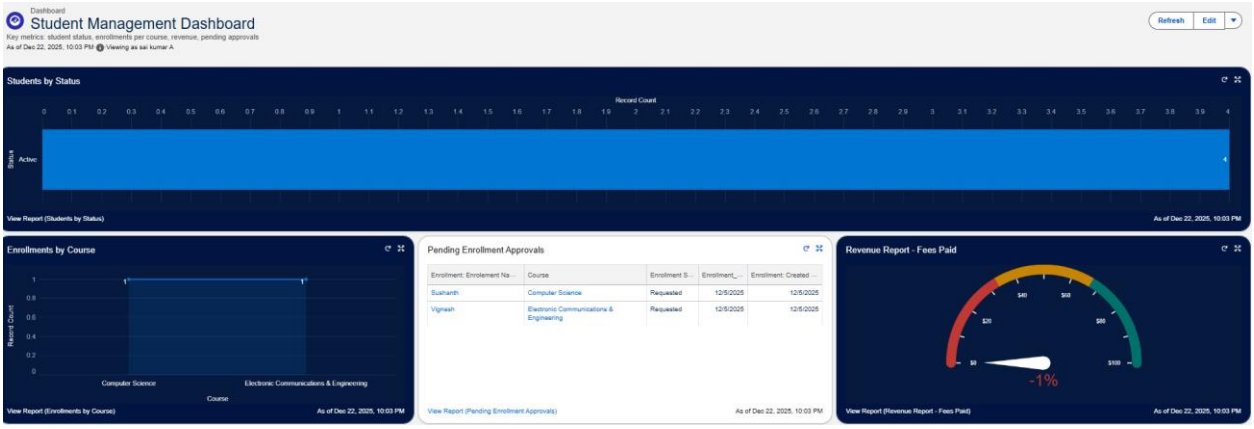


Activity 1: Student Management Dashboard

- Dashboards → New Dashboard → Name: **Student Management Dashboard**
- Donut Chart using Students by Status report



- Bar Chart using Enrollments by Course report
- Gauge using Revenue Report – Fees Paid (sum of revenue field)
- Table using Pending Enrollment Approvals report



- Arrange components on canvas → Save → Done

Activity 1: Configure Profiles

Admin	Full access
Enrollment Officer	Create / Update Enrollments
Instructor	Read-only access to assigned Enrollments

For the Enrollment Officer profile: Enrollment__c set to Read, Create, Edit; Student__c / Course__c set to Read (and Edit if required). The Instructor profile is configured similarly with Read-Only access on Enrollments.

Activity 2: Create Permission Sets

- Custom Object Access — additional access to project objects if needed
- Dashboard Editor — Create/Customize Reports & Dashboards permissions
- Assign to relevant users

Activity 3: Configure Sharing & OWD

Student__c	Public Read/Write
Course__c	Public Read/Write
Enrollment__c	Private

For Instructor access to only their assigned Enrollments, a User lookup can be added on Enrollment (Instructor User), with a simple Record-Triggered Flow setting OwnerId = Instructor User so instructors see only their own records.

7. Functional & Performance Testing

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features of SECMS. These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; verification of record type behavior for different course categories; flow execution for automated enrollment approvals and seat availability checks; and Apex trigger functionality to ensure automatic confirmation and record updates. Screenshots of generated reports and dashboards were also documented to validate accurate data visualization, enrollment tracking, and performance monitoring.

8. Advantages & Disadvantages

Centralized data	Salesforce dependency

Automated workflows	Requires initial configuration effort
Real-time reporting	

9. Conclusion

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle — from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce's automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. Future Scope

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. Appendix

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You